EXPERIMENT NO 1

Design a professional-looking Professional Resume or research paper in LaTeX, using Overleaf/Latex, with proper formatting, sections, and a table of contents.

Aim

To design a professional-looking resume using LaTeX with proper formatting and structured sections.

Objective

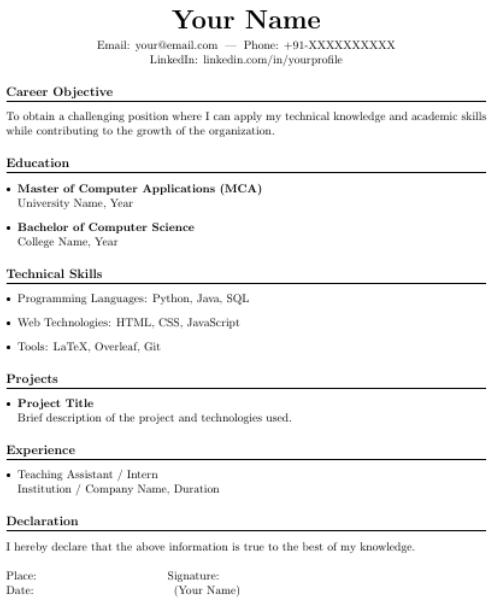
- To understand the structure of a professional resume in LaTeX.
- To learn the use of document classes, packages, and formatting commands.
- To create well-organized sections such as education, skills, and experience.
- To develop a clean, consistent, and professional resume layout suitable for academic and industry use.

```
\documentclass[11pt,a4paper]{article}
% Packages
\usepackage{geometry}
\geometry{margin=1in}
\usepackage{titlesec}
\usepackage{enumitem}
\usepackage{hyperref}
% Section formatting
\titleformat{\section}{\large\bfseries}{}{0em}{}
[\titlerule]
\begin{document}
    \begin{center}
        {\Huge \textbf{Your Name}}\\
        \vspace{0.2cm}
        Email: your@email.com \;|\; Phone: +91-
XXXXXXXXXX\\
        LinkedIn: linkedin.com/in/yourprofile
    \end{center}
    \section*{Career Objective}
    To obtain a challenging position where I can
apply my technical knowledge and academic skills
while contributing to the growth of the organization.
    \section*{Education}
    \begin{itemize}[leftmargin=*]
        \item \textbf{Master of Computer
Applications (MCA)}\\
        University Name, Year
        \item \textbf{Bachelor of Computer
Science}\\
        College Name, Year
    \end{itemize}
    \section*{Technical Skills}
    \begin{itemize}[leftmargin=*]
        \item Programming Languages: Python,
Java, SQL
        \item Web Technologies: HTML, CSS,
```

JavaScript

```
        \item Tools: LaTeX, Overleaf, Git
    \end{itemize}
    \section*{Projects}
    \begin{itemize}[leftmargin=*]
        \item \textbf{Project Title}\\
        Brief description of the project and
technologies used.
    \end{itemize}
    \section*{Experience}
    \begin{itemize}[leftmargin=*]
        \item Teaching Assistant / Intern\\
        Institution / Company Name, Duration
    \end{itemize}
    \section*{Declaration}
    I hereby declare that the above information is
true to the best of my knowledge.
    \vspace{0.5cm}
    \noindent Place: \hspace{4cm} Signature:\\
    Date: \hspace{4.3cm} (Your Name)
\end{document}\documentclass[11pt,a4paper]
{article}
```

OUTPUT:



Your Name

Email: your@email.com — Phone: +91-XXXXXXXXXX
LinkedIn: linkedin.com/in/yourprofile

Career Objective

To obtain a challenging position where I can apply my technical knowledge and academic skills while contributing to the growth of the organization.

Education

- **Master of Computer Applications (MCA)**
  University Name, Year
- **Bachelor of Computer Science**
  College Name, Year

Technical Skills

- Programming Languages: Python, Java, SQL
- Web Technologies: HTML, CSS, JavaScript
- Tools: LaTeX, Overleaf, Git

Projects

- **Project Title**
  Brief description of the project and technologies used.

Experience

- Teaching Assistant / Intern
  Institution / Company Name, Duration

Declaration

I hereby declare that the above information is true to the best of my knowledge.

Place: Signature:
Date: (Your Name)

Conclusion

A professional resume was successfully designed using LaTeX. This practical demonstrated how LaTeX helps in creating clean, well-structured, and professional resumes with consistent formatting, making it an effective tool for academic and career-oriented documentation.

EXPERIMENT NO 2:

Create a presentation in Overleaf / LaTeX, focusing on layout, image integration, and slide transitions.

Aim

To create a professional presentation using Overleaf / LaTeX (Beamer) by focusing on slide layout, image integration, and slide transitions.

Objectives

1. To understand the basic structure of a Beamer presentation in LaTeX.
2. To design well-organized slides using layouts such as titles, lists, and columns.
3. To insert and format images properly within presentation slides.
4. To apply slide transitions and effects to improve visual presentation.
5. To develop a clean and professional presentation using Overleaf.

Code:

```
\documentclass[aspectratio=169]{beamer}
```

```
% ------------------ Theme & Colors ------------------
\usetheme{Madrid}
\usecolortheme{default}

% ------------------ Packages ------------------
\usepackage{graphicx}   % For images
\usepackage{booktabs}   % For tables
\usepackage{multicol}   % Multiple columns
\usepackage{hyperref}   % Links
\usepackage{animate}   % Animations
\usepackage{tikz}       % Diagrams

% ------------------ Slide Transitions ------------------
\usepackage{pdfpages}
\setbeamertemplate{navigation symbols}{}

% Add transition effect
\usepackage{etoolbox}
\AtBeginSection[]{
    \begin{frame}[plain]
        \transfade
        \centering
        \Huge \insertsection
    \end{frame}
}
```

```
% ------------------ Title Information ------------------
\title{Introduction to LaTeX Beamer}
\subtitle{Layout, Images, and Transitions}
\author{Bhagyashree}
\institute{Department of Computer Science}
\date{\today}

% ------------------ Document Start ------------------
\begin{document}

    % ------------------ Title Slide ------------------
    \begin{frame}
        \transfade
        \titlepage
    \end{frame}

    % ------------------ Outline Slide ------------------
    \begin{frame}{Outline}
        \transwipe[direction=90]
        \tableofcontents
    \end{frame}

    % ------------------ Section 1 ------------------
    \section{Introduction}

    \begin{frame}{What is Beamer?}
```

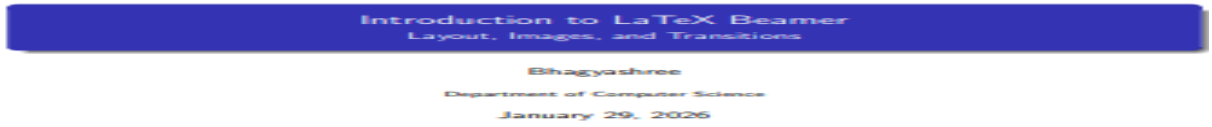
```
    \transfade
    \begin{itemize}
        \item Beamer is a LaTeX class for presentations
        \item Produces professional-quality slides
        \item Widely used in academia and research
    \end{itemize}
\end{frame}

% ------------------ Section 2 ------------------
\section{Layout Design}

\begin{frame}{Two-Column Layout}
    \transwipe[direction=0]
    \begin{columns}
        \column{0.5\textwidth}
        \begin{itemize}
            \item Structured layout
            \item Easy alignment
            \item Professional look
        \end{itemize}

        \column{0.5\textwidth}
        \begin{block}{Tip}
            Use columns to balance text and visuals.
        \end{block}
```

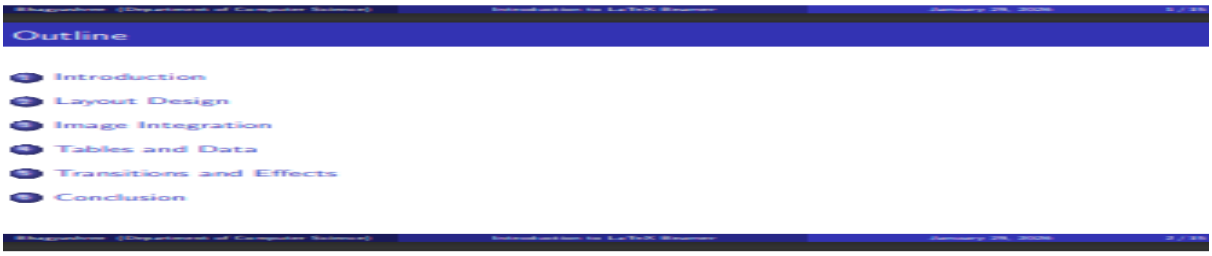
```
    \end{columns}
\end{frame}

% ------------------ Section 3 ------------------
\section{Image Integration}

\begin{frame}{Including Images}
    \transfade
    \begin{figure}
        \centering
        \includegraphics[width=0.7\textwidth]{example-image}
        \caption{Sample Image in Beamer}
    \end{figure}
\end{frame}

% ------------------ Section 4 ------------------
\section{Tables and Data}

\begin{frame}{Sample Table}
    \transwipe[direction=180]
    \begin{table}
        \centering
        \begin{tabular}{lcc}
            \toprule
            Feature & Supported & Easy to Use \\
```

```
                \midrule
                Layout & Yes & Yes \\
                Images & Yes & Yes \\
                Transitions & Yes & Moderate \\
                \bottomrule
            \end{tabular}
            \caption{Beamer Features}
        \end{table}
    \end{frame}

    % ------------------ Section 5 ------------------
    \section{Transitions and Effects}

    \begin{frame}{Slide Transitions}
        \transfade
        \begin{itemize}
            \item \texttt{\textbackslash transfade}
            \item \texttt{\textbackslash transwipe}
            \item \texttt{\textbackslash transblindshorizontal}
        \end{itemize}
    \end{frame}

    % ------------------ Section 6 ------------------
    \section{Conclusion}
```

```latex
    \begin{frame}{Conclusion}
        \transfade
        \begin{enumerate}
            \item Beamer creates professional presentations
            \item Layouts and images are easy to manage
            \item Transitions enhance visual appeal
        \end{enumerate}
    \end{frame}

    % ------------------ Thank You Slide ------------------
    \begin{frame}[plain]
        \transfade
        \centering
        \Huge Thank You!\\
        \vspace{0.5cm}
        \Large Questions?
    \end{frame}

\end{document}
```

Output:





Conclusion

In this practical, a presentation was successfully created using Overleaf and LaTeX Beamer. Different slide layouts were designed, images were integrated effectively, and slide transition effects were applied. This demonstrates that LaTeX Beamer is a powerful tool for creating professional and structured presentations suitable for academic and technical use.

EXPERIMENT NO 3

Installation Of Linux Operating System (Using Windows Subsystem For Linux – Wsl)

Aim:

To install the Linux Operating System (Ubuntu) on Windows using Windows Subsystem for Linux (WSL).

Requirements:

- Windows 10/11 (64-bit)
- Internet Connection
- Administrator Access

Procedure:

Step 1: Enable Required Windows Features

1. Open Windows Search.
2. Type: Turn Windows features on or off
3. Enable the following:
   - Virtual Machine Platform
   - Windows Subsystem for Linux
4. Click OK.
5. Click Restart Now.

Step 2: Check WSL Status

1. Open Command Prompt (Run as Administrator).
2. Type:
3. wsl --status

   This shows WSL version and status.

Step 3: Update WSL

In Command Prompt, type:

wsl --update

This updates the WSL kernel.

Step 4: Install Ubuntu from Microsoft Store

1. Open Microsoft Store.
2. Search: Ubuntu
3. Click Download / Get
4. Wait for installation to complete.

5. Click Open

Step 5: Create UNIX User Account

When Ubuntu opens for the first time:

1. Enter UNIX username
2. Enter Password
3. Confirm Password

(Note: Password will not be visible while typing.)

Step 6: Verify Installation

After login, type:

```
ls
pwd
cd /
ls
```

If directories like bin, etc, home, usr, var appear Installation Successful

Sample Output (Terminal View)

You will see:

- Ubuntu version details
- System information
- Linux directory structure

Result/ conclusion:

Ubuntu Linux was successfully installed on Windows using Windows Subsystem for Linux (WSL).

EXPERIMENT NO 4

Perform basic file operations using Linux command-line tools, such as creating, editing, moving, and deleting files and directories.

Aim:

To perform basic file operations in Linux using command-line tools such as creating, editing, copying, moving, and deleting files and directories.

Theory:

Linux is a multi-user, multi-tasking operating system that provides powerful command-line tools for file management. The Linux file system is hierarchical in structure and allows users to efficiently manage files and directories using various commands.

Some commonly used Linux file operation commands include:

- touch – Creates empty files
- mkdir – Creates directories
- ls – Lists files and directories
- cat – Displays file content
- nano / vi – Edits files
- cp – Copies files and directories
- mv – Moves or renames files
- rm – Deletes files
- rmdir – Removes empty directories

These commands help users manage data efficiently in a Linux environment.

Procedure:

1. Create a Directory

mkdir lab

2. Navigate into Directory

cd lab

3. Create a File

touch test.txt

4. Edit the File

nano test.txt

(Type some content and save using Ctrl + X, then Y, then Enter)

5. Display File Content

cat test.txt

6. Copy the File

cp test.txt copy.txt

7. Create Another Directory

mkdir folder1

8. Move File to Directory

mv copy.txt folder1/

9. Delete a File

rm test.txt

10. Remove Directory

rm -r folder1

11. Go Back and Remove Main Directory

```
cd ..
rmdir lab
```

Commands Used:

Command	Description
mkdir	Create directory
cd	Change directory
touch	Create empty file
nano	Edit file
cat	Display file content
cp	Copy file
mv	Move/Rename file
rm	Delete file
rmdir	Remove empty directory

Result/ conclusion :

In this experiment, we successfully performed basic file operations in Linux using command-line tools. We learned how to create, edit, copy, move, and delete files and directories, which are essential skills for working in a Linux environment. Basic file operations such as creating, editing, copying, moving, and deleting files and directories were successfully performed using Linux command-line tools.

EXPERIMENT NO 5

Create a basic responsive web page using Bootstrap, incorporating various Bootstrap components like navigation bars, buttons, and forms.

Aim:Create a basic responsive web page using Bootstrap, incorporating various Bootstrap components like navigation bars, buttons, and forms.

Prerequisites:

- Basic understanding of HTML and CSS.
- Knowledge of the Bootstrap framework.
- A code editor (e.g., VS Code, Sublime Text, or Notepad++).
- A web browser (e.g., Google Chrome, Firefox).
- Internet connection to access Bootstrap CDN.

Theory:Bootstrap is a popular open-source front-end framework developed by Twitter to help developers create responsive and mobile-first web pages quickly and efficiently. It provides a collection of pre-designed CSS and JavaScript components that simplify the development process, ensuring a consistent and professional look across different devices and screen sizes.

Key Features of Bootstrap:

1. Responsive Grid System.
2. Predefined Components.
3. Customizable Themes and Utilities.
4. Mobile-First Approach.
5. JavaScript Plugins.

Procedure:

Step 1: Setting Up the Environment

1. Open your code editor and create a new HTML file (e.g., index.html).

2.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Bootstrap Web Page</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/
```

Include the Bootstrap CDN in the <head> section of your HTML file:

Step 2: Creating a Navigation Bar

1. Add the following code inside the <body> tag to create a responsive navigation bar:

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
<div class="container-fluid">
<a class="navbar-brand" href="#">My Website</a>
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
<span class="navbar-toggler-icon"></span>
</button>
```

Step 3: Adding a Jumbotron Section

1.

```
<div class="container mt-4">
<div class="p-5 bg-light rounded">
<h1>Welcome to My Website</h1>
<p class="lead">This is a simple responsive webpage using Bootstrap
components.</p>
<a class="btn btn-primary" href="#">Learn More</a>
</div>
```

Create a welcoming section using Bootstrap’s jumbotron-style container:

Step 4: Adding a Form

1.

```
<div class="container mt-4">
<h2>Contact Us</h2>
<form>
<div class="mb-3">
<label for="name" class="form-label">Name</label>
<input type="text" class="form-control" id="name" placeholder="Enter your name">
</div>
```

Create a simple contact form using Bootstrap's form components:

Step 5: Adding a Footer

1.

```
<footer class="bg-dark text-white text-center py-3 mt-4">
<p>&copy; 2025 My Website. All Rights Reserved.</p>
</footer>
```

Include a footer section to complete the layout:

Step 6: Linking Bootstrap JavaScript

1.

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/
bootstrap.bundle.min.js"></script>
</body>
</html>
```

Include the Bootstrap JavaScript CDN before closing the </body> tag:

Expected Outcome: Upon completing this lab, you should have a fully functional and responsive web page that includes a navigation bar, a jumbotron-style introduction, a contact form, and a footer. The page should adjust dynamically based on the screen size.

Conclusion: In this lab, we have successfully built a basic responsive web page using Bootstrap. We incorporated fundamental Bootstrap components such as a navbar, buttons, forms, and containers to create a structured and user-friendly layout.

EXPERIMENT NO: 6

Develop a Small Web Application with a User Login Page Using Bootstrap, Demonstrating Form Handling and Responsive Design

---

Aim:Develop a Small Web Application with a User Login Page Using Bootstrap, Demonstrating Form Handling and Responsive Design

Objective: To develop a small web application featuring a user login page using Bootstrap. The application will demonstrate form handling and responsive design principles.

Prerequisites:

- Basic knowledge of HTML, CSS, and JavaScript.
- Understanding of Bootstrap framework.
- Familiarity with form validation and handling user input.
- A code editor (e.g., VS Code, Sublime Text, or Notepad++).
- A web browser (e.g., Google Chrome, Firefox).
- Internet connection to access Bootstrap CDN.

Theory:

Bootstrap is a front-end framework that facilitates the development of responsive and

mobile-friendly web applications. It includes a grid system, pre-designed UI components, and JavaScript plugins that simplify the creation of modern web pages.

A user login page is a critical component of web applications, allowing users to securely authenticate and access personalized content. Form handling in web applications ensures that user inputs are validated and submitted correctly, improving security and usability.

Key concepts demonstrated in this practical:

1. Responsive Design: Ensuring the login page adapts to different screen sizes.
2. Bootstrap Forms: Using Bootstrap classes for styling and structuring forms.
3. Form Validation: Implementing client-side validation for secure user input.
4. Navigation & Layout: Utilizing Bootstrap components for a consistent UI.

Procedure:

Step 1: Setting Up the Environment

1. Open your code editor and create a new HTML file (e.g., login.html).
2.
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login Page</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/
```

   Include the Bootstrap CDN in the <head> section:

Step 2: Creating a Navigation Bar

1.
```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
<div class="container-fluid">
<a class="navbar-brand" href="#">Data Science App</a>
</div>
</nav>
```

   Add a navigation bar to the page:

```
<div class="container d-flex justify-content-center align-items-center vh-100">
<div class="card p-4 shadow-sm" style="width: 350px;">
<h3 class="text-center">Login</h3>
<form>
<div class="mb-3">
<label for="email" class="form-label">Email Address</label>
<input type="email" class="form-control" id="email" placeholder="Enter your email" required>
</div>
<div class="mb-3">
<label for="password" class="form-label">Password</label>
<input type="password" class="form-control" id="password" placeholder="Enter your
password" required>
</div>
<button type="submit" class="btnbtn-primary w-100">Login</button>
</form>
</div>
</div>
```

Step 3: Creating the Login Form

1. Add a Bootstrap-styled login form:

Step 4: Adding Basic Form Validation

1. Include JavaScript for client-side validation:

```
<script>
document.querySelector("form").addEventListener("submit", function(event) {
        let email = document.getElementById("email").value;
        let password = document.getElementById("password").value;
        if (email === "" || password === "") {
alert("All fields are required!");
event.preventDefault();
        }
    });
```

Step 5: Linking Bootstrap JavaScript

1.

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/
bootstrap.bundle.min.js"></script>
</body>
</html>
```

Include Bootstrap's JavaScript library before closing the </body> tag:

Expected Outcome: Upon completing this practical, you will have a fully functional and responsive login page that adapts to different screen sizes and includes basic form validation.

Conclusion: In this practical, we developed a user login page using Bootstrap. We incorporated Bootstrap form components, navigation bars, and responsive design principles to ensure usability and accessibility.

EXPERIMENT NO : 7

Setting Up and Configuring MATLAB Environment

Aim

To install, set up, and configure the MATLAB environment and verify its proper functioning.

Objectives

1. To install MATLAB software.
2. To understand the MATLAB desktop interface.
3. To configure working directory and path settings.
4. To install required toolboxes.
5. To execute a basic MATLAB program.

Software Required

- MATLAB (Latest Version)
- Windows / Linux Operating System
- Internet Connection

Theory

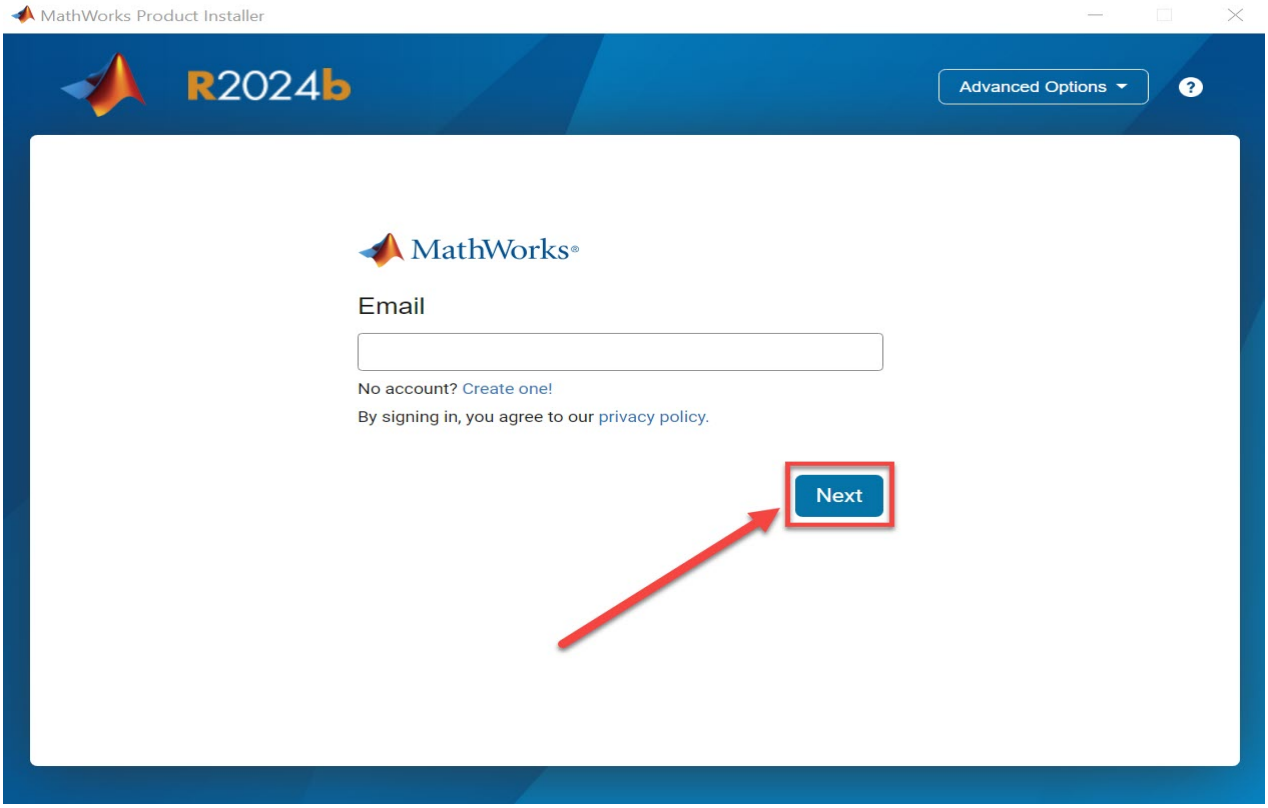
MATLAB (Matrix Laboratory) is a high-level programming language and numerical computing environment developed by MathWorks. It is widely used for data analysis, algorithm development, simulation, modeling, and visualization.

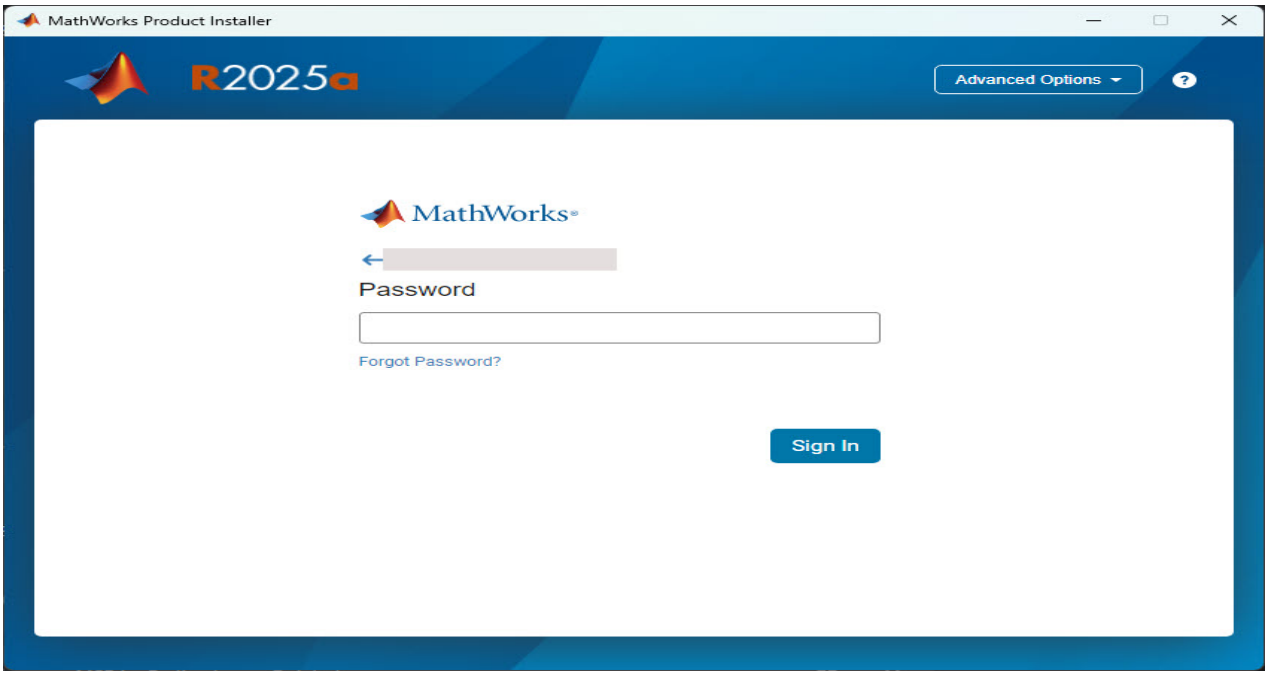
The MATLAB desktop consists of:

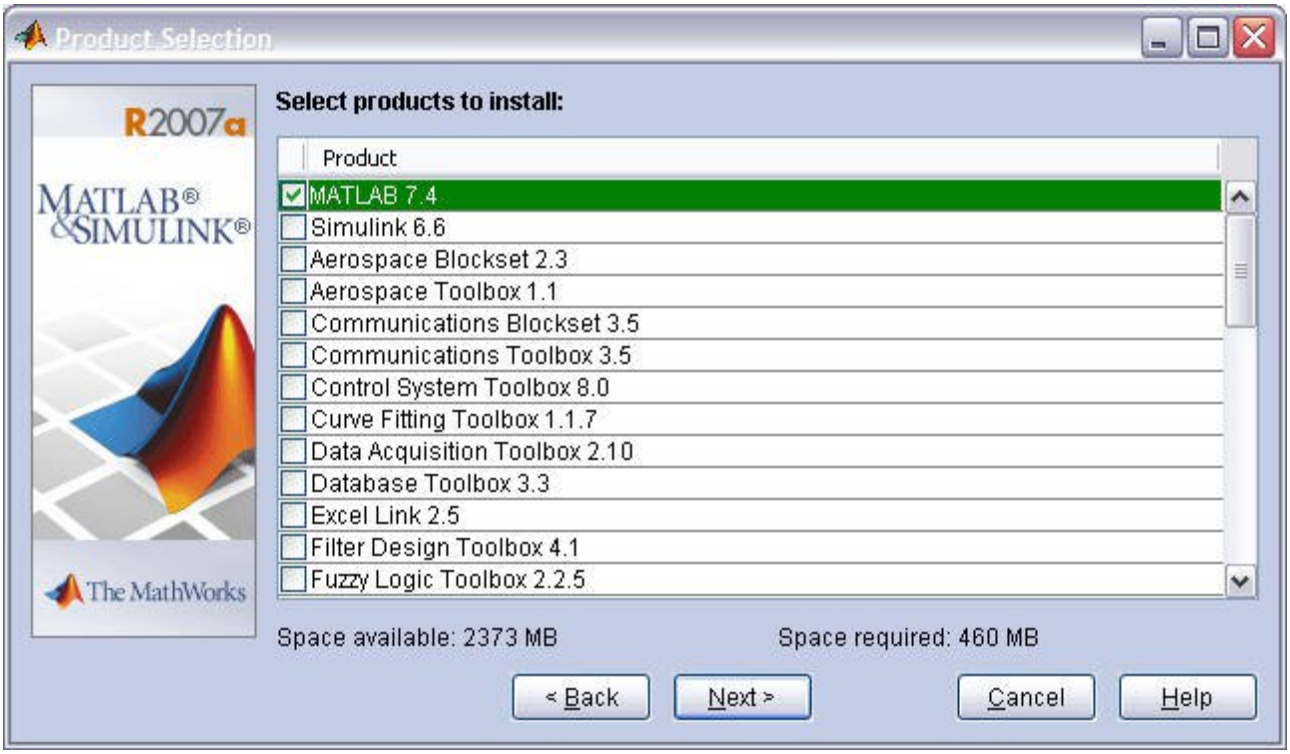
- Command Window
- Workspace
- Current Folder
- Editor
- Command History

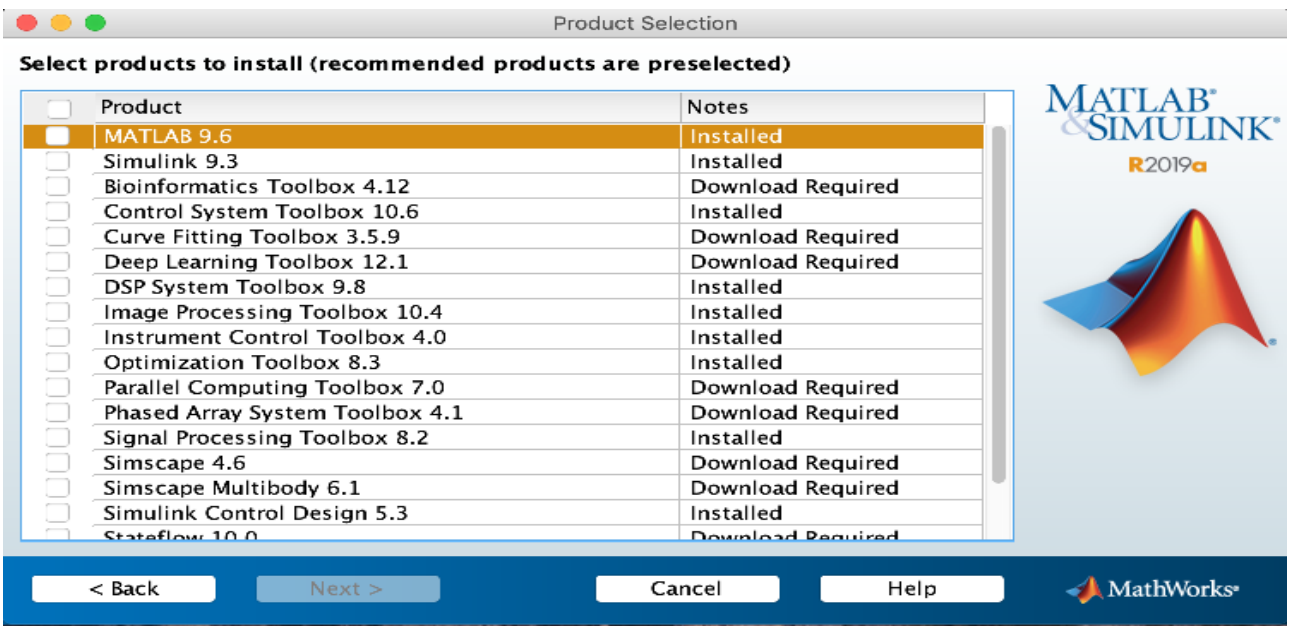
Procedure

Step 1: Install MATLAB


MathWorks Product Installer
R2024b
Advanced Options
MathWorks
Email
No account? Create one!
By signing in, you agree to our privacy policy.
Next


MathWorks Product Installer
R2025a
Advanced Options
MathWorks
Password
Forgot Password?
Sign In

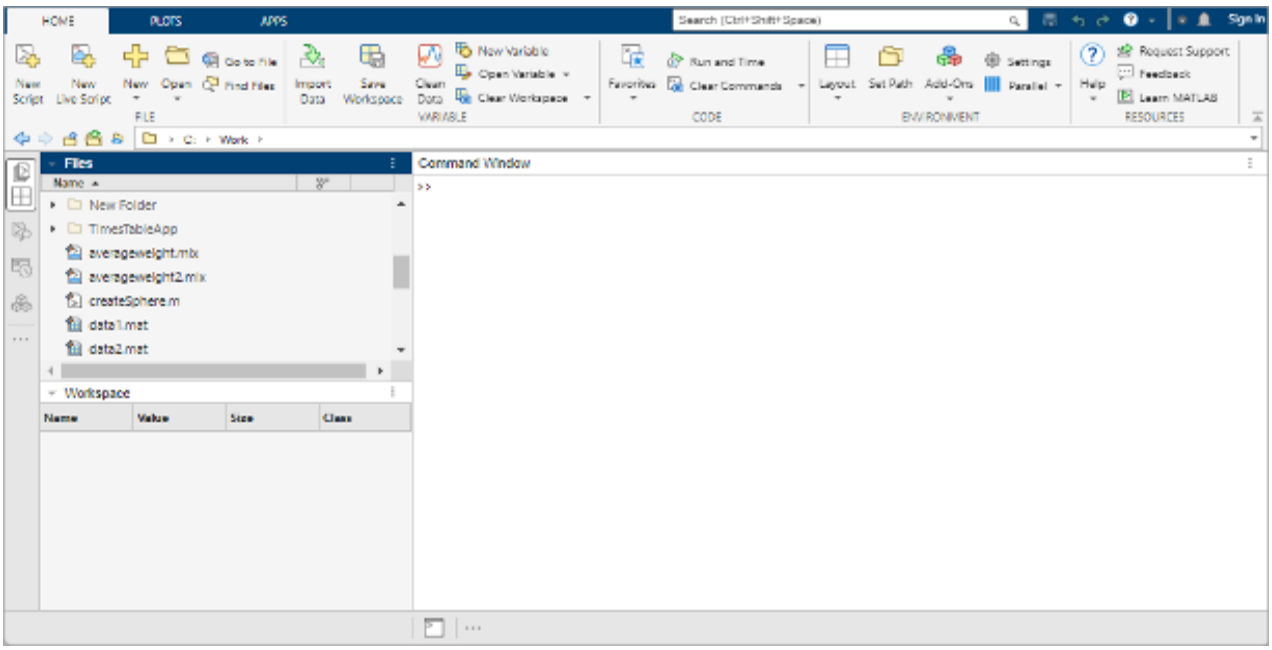


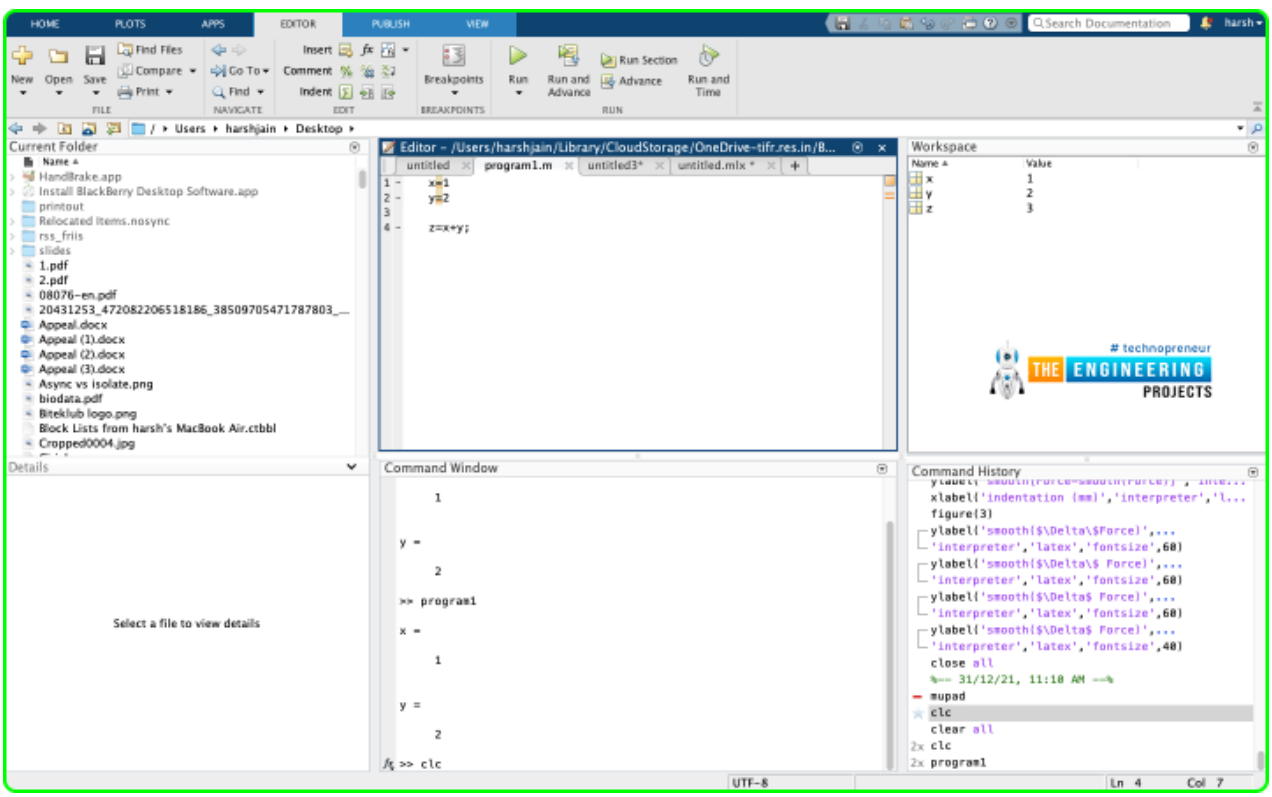


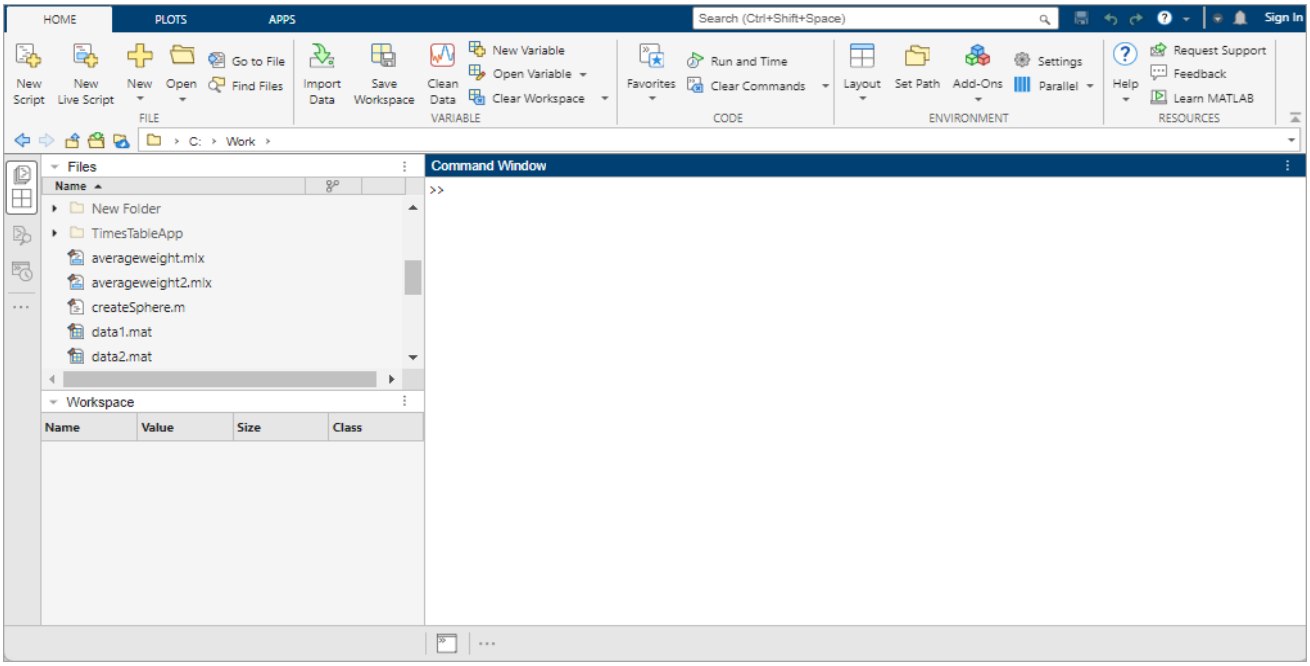
1. Visit the official website: https://www.mathworks.com
2. Sign in with your MathWorks account.
3. Download the MATLAB installer.
4. Run the installer.
5. Select license type (Student/Trial/Institution).
6. Choose required toolboxes.

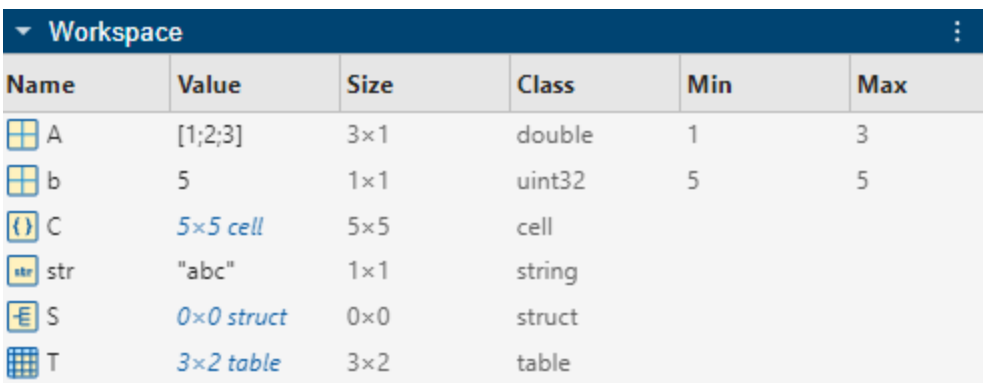
7. Select installation folder.
8. Click Install and wait for completion.

Step 2: Launch MATLAB









1. Open MATLAB from Start Menu.
2. Observe the MATLAB Desktop interface.
3. Identify:
   - o Command Window
   - o Workspace
   - o Current Folder
   - o Editor

Step 3: Set Working Directory

1. Click on Current Folder panel.
2. Select your project folder.

OR

Use command:

cd 'C:\MATLAB_Project'

Step 4: Add Folder to Path

1. Go to Home Set Path Add Folder
2. Click Save

OR

```
addpath('C:\MATLAB_Project')
savepath
```

Step 5: Install Toolboxes

1. Go to Home Add-Ons Get Add-Ons
2. Search for required toolbox
3. Click Install

Example Toolboxes:

- Image Processing Toolbox
- Signal Processing Toolbox
- Deep Learning Toolbox

Step 6: Execute Test Program

In Command Window type:

```
a = 10;
b = 20;
c = a + b
```

Output:

```
c = 30
```

Step 7: Create and Run Script

1. Click New Script
2. Type:

```
disp('Hello MATLAB')
```

3. Click Run

Output:

```
Hello MATLAB
```

Result

MATLAB was successfully installed and configured. Basic commands and script execution were performed successfully.

Precautions

1. Ensure stable internet connection during installation.
2. Select correct license type.

3. Save path after adding folders.
4. Verify toolbox compatibility before installation.

Experiment No:8

Implement a simple MATLAB script to perform matrix operations (addition, subtraction, multiplication) and display the results.

Aim

To implement a simple MATLAB script to perform matrix addition, subtraction, and multiplication and display the results.

Objectives

1. To create matrices in MATLAB.
2. To perform matrix addition.
3. To perform matrix subtraction.
4. To perform matrix multiplication.
5. To display results using MATLAB commands.

Theory

In MATLAB, matrices are fundamental data structures.

If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

Then:

- Addition: (A + B)
- Subtraction: (A - B)
- Multiplication: (A \times B) (Row × Column rule)

Note: For addition and subtraction, matrices must be of the same size.

For multiplication, number of columns of first matrix = number of rows of second matrix.

Procedure

Step 1: Open MATLAB

Launch MATLAB from the Start Menu.

Step 2: Create a New Script

Click Home New Script

Step 3: Write the Following Code

```
% Matrix Operations Script

clc;
clear;

% Define matrices
A = [1 2; 3 4];
B = [5 6; 7 8];

% Matrix Addition
C = A + B;

% Matrix Subtraction
D = A - B;

% Matrix Multiplication
E = A * B;

% Display Results
disp('Matrix A:')
disp(A)

disp('Matrix B:')
disp(B)

disp('Addition (A + B):')
disp(C)

disp('Subtraction (A - B):')
disp(D)

disp('Multiplication (A * B):')
disp(E)
```

Step 4: Run the Script

Click Run button.

Expected Output

```
Matrix A:
   1   2
   3   4

Matrix B:
   5   6
   7   8

Addition (A + B):
   6   8
```

```
   10   12

Subtraction (A - B):
   -4   -4
   -4   -4

Multiplication (A * B):
   19   22
   43   50
```

Result

Matrix addition, subtraction, and multiplication were successfully performed using MATLAB script, and the results were displayed correctly.

Precautions

1. Ensure matrices have same dimensions for addition and subtraction.
2. Check compatibility condition for multiplication.
3. Use * for matrix multiplication (not .* unless performing element-wise multiplication).

Experiment No: 9

Setting Up and Configuring Two Different Hypervisors

Aim

To install, set up, and configure two different hypervisors — Oracle VM VirtualBox and VMware Workstation — and create virtual machines to understand virtualization concepts.

Objectives

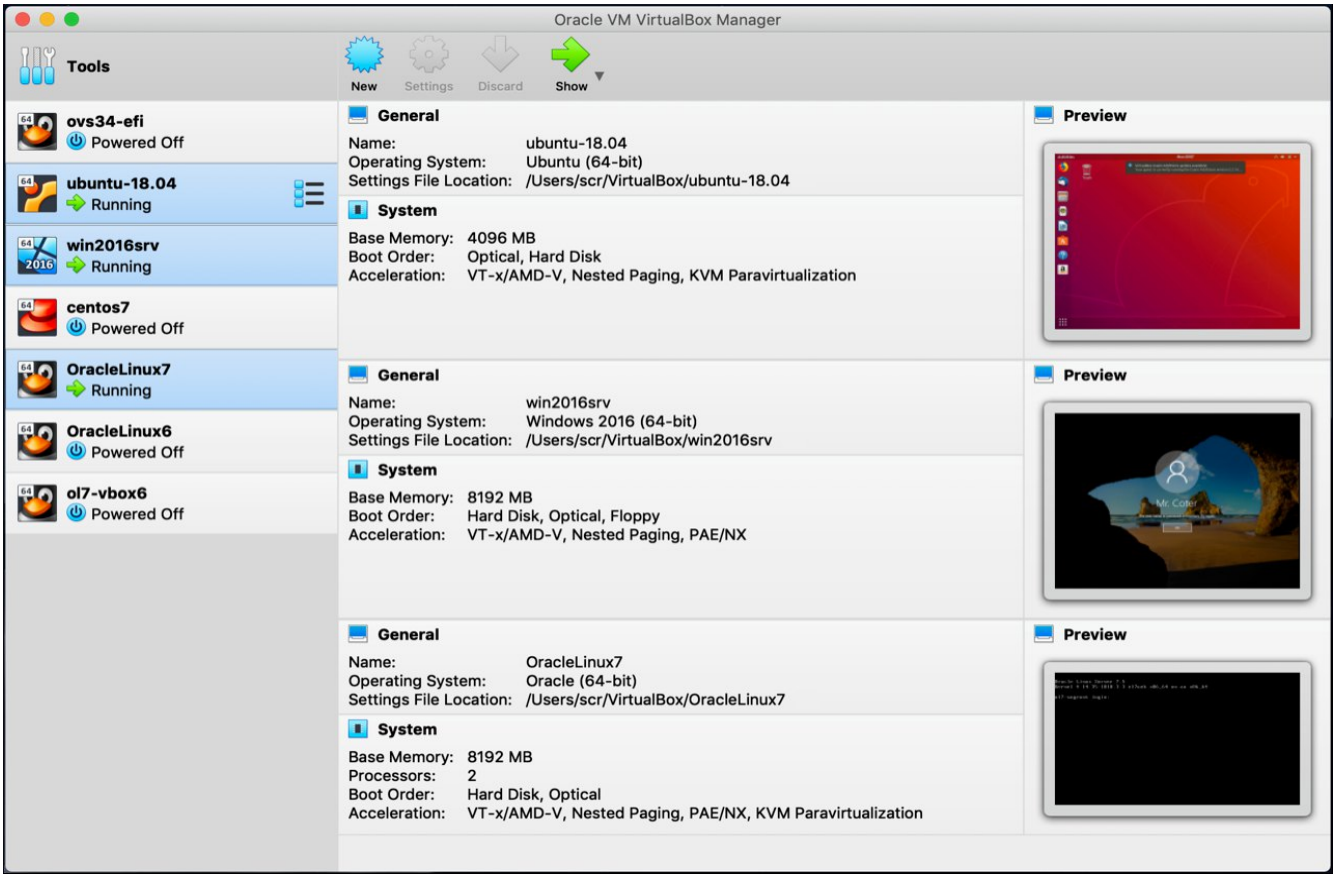
1. To understand the concept of virtualization and hypervisors.
2. To install and configure VirtualBox on a Windows system.
3. To install and configure VMware Workstation on a Windows system.
4. To create and run a virtual machine using Ubuntu Linux ISO.
5. To configure VM settings such as RAM, CPU, storage, and networking.
6. To compare features and performance of both hypervisors.

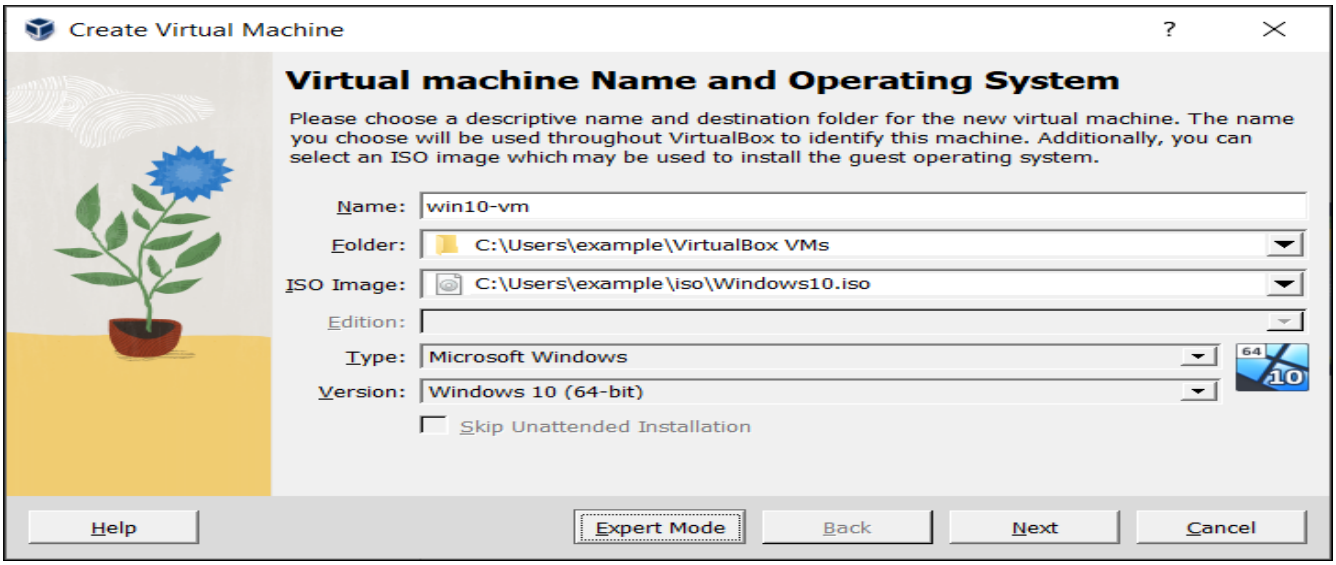
Tools & Software Required

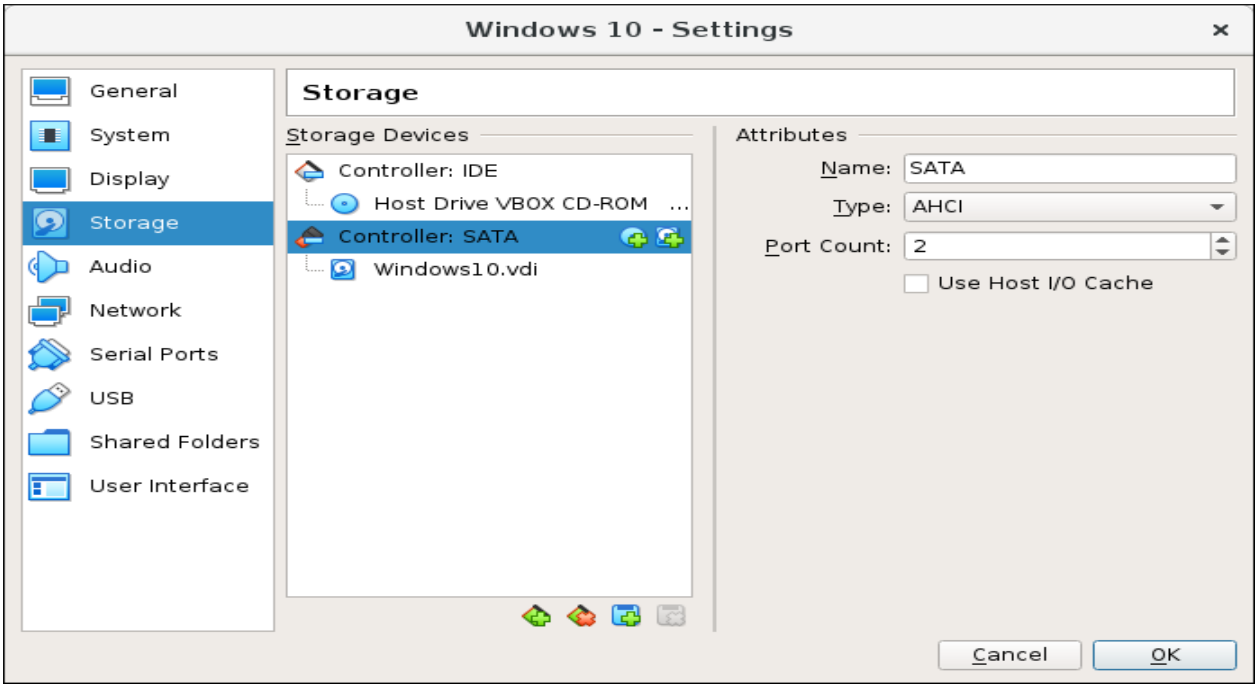
- Windows 10/11 Operating System
- Ubuntu ISO File
- Oracle VM VirtualBox
- VMware Workstation Player

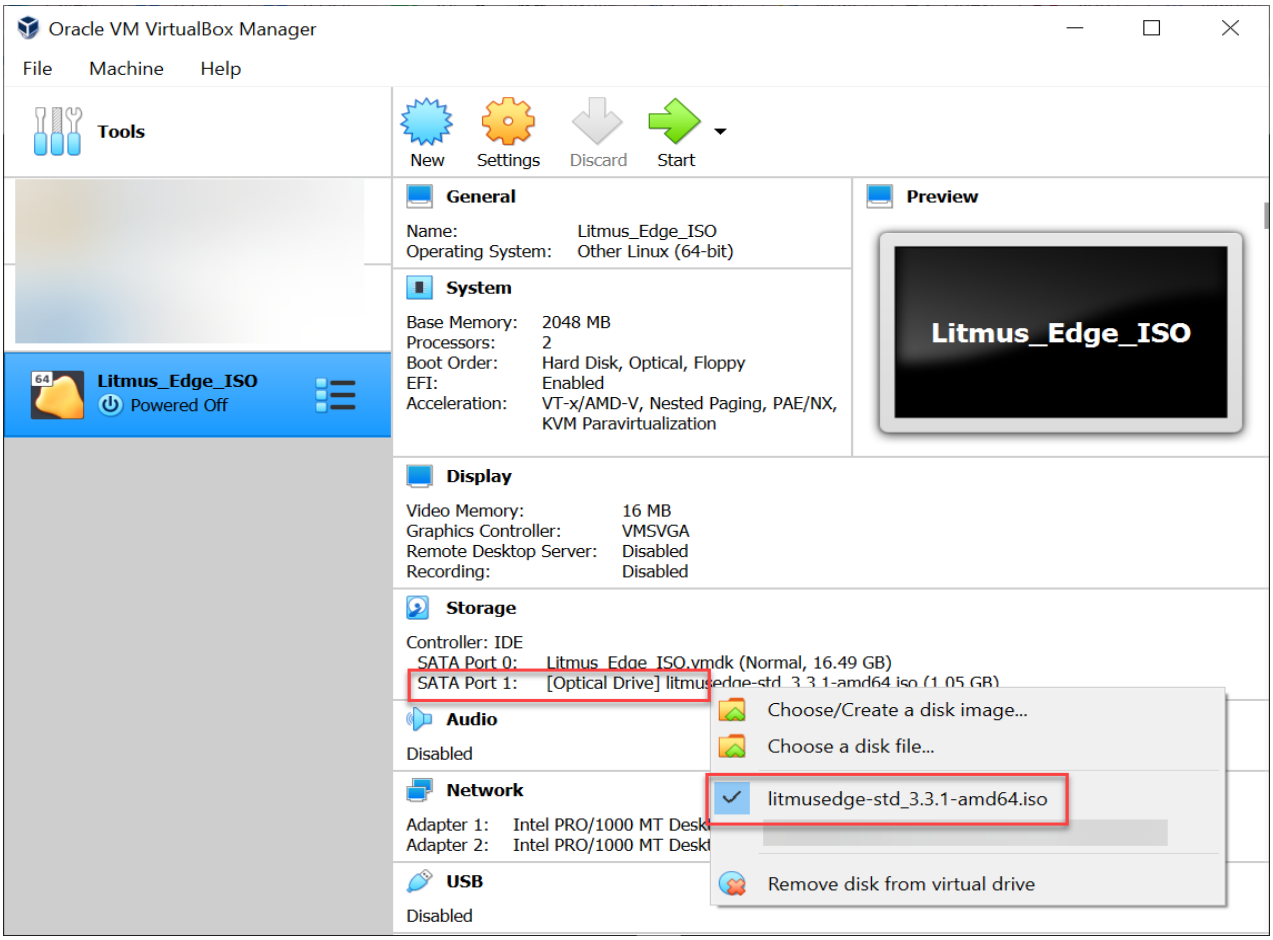
Procedure

Part A: Installation and Configuration of VirtualBox


Oracle VM VirtualBox Manager
Tools
New
Settings
Discard
Show
ovs34-efi
Powered Off
ubuntu-18.04
Running
win2016srv
Running
centos7
Powered Off
OracleLinux7
Running
OracleLinux6
Powered Off
ol7-vbox6
Powered Off
General
Name: ubuntu-18.04
Operating System: Ubuntu (64-bit)
Settings File Location: /Users/scr/VirtualBox/ubuntu-18.04
System
Base Memory: 4096 MB
Boot Order: Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM Paravirtualization
Preview
General
Name: win2016srv
Operating System: Windows 2016 (64-bit)
Settings File Location: /Users/scr/VirtualBox/win2016srv
System
Base Memory: 8192 MB
Boot Order: Hard Disk, Optical, Floppy
Acceleration: VT-x/AMD-V, Nested Paging, PAE/NX
Preview
General
Name: OracleLinux7
Operating System: Oracle (64-bit)
Settings File Location: /Users/scr/VirtualBox/OracleLinux7
System
Base Memory: 8192 MB
Processors: 2
Boot Order: Hard Disk, Optical
Acceleration: VT-x/AMD-V, Nested Paging, PAE/NX, KVM Paravirtualization
Preview


Create Virtual Machine
Virtual machine Name and Operating System
Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.
Name: win10-vm
Folder: C:\Users\example\VirtualBox VMs
ISO Image: C:\Users\example\iso\Windows10.iso
Edition:
Type: Microsoft Windows
Version: Windows 10 (64-bit)
Skip Unattended Installation
Help
Expert Mode
Back
Next
Cancel

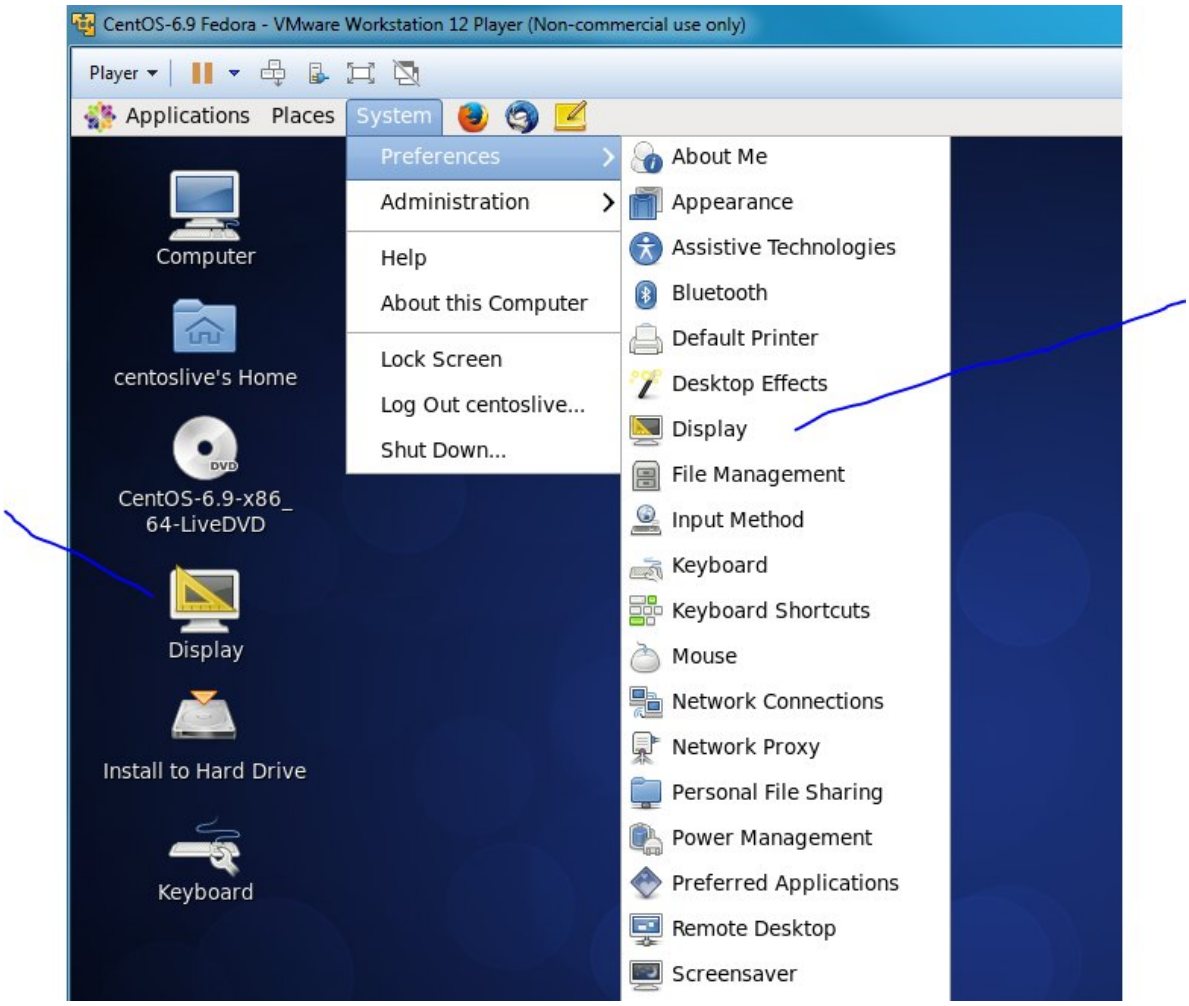


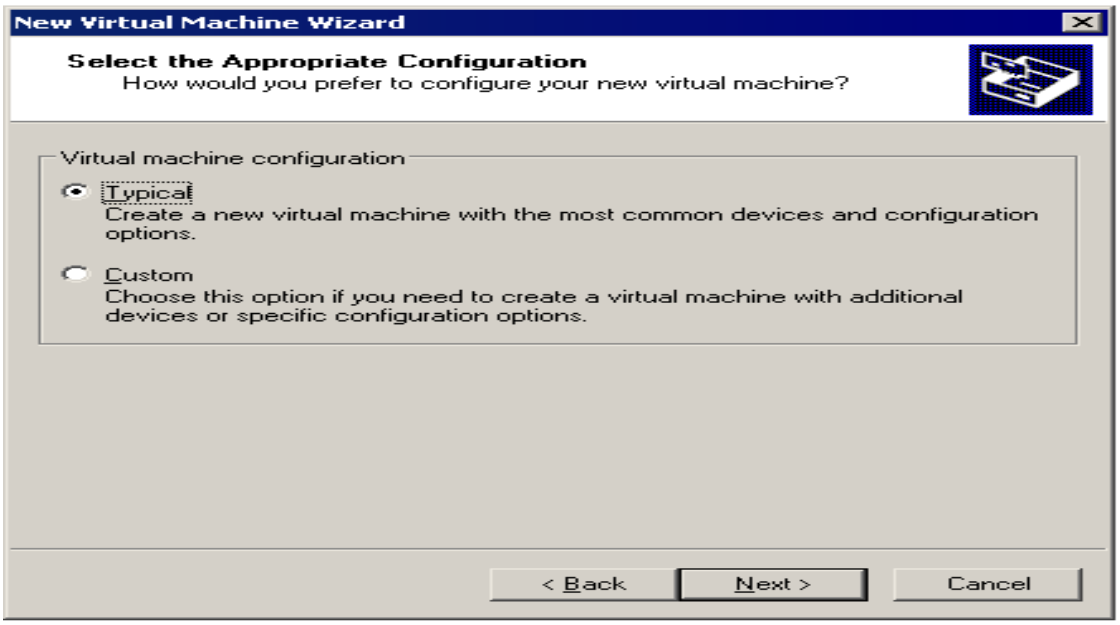


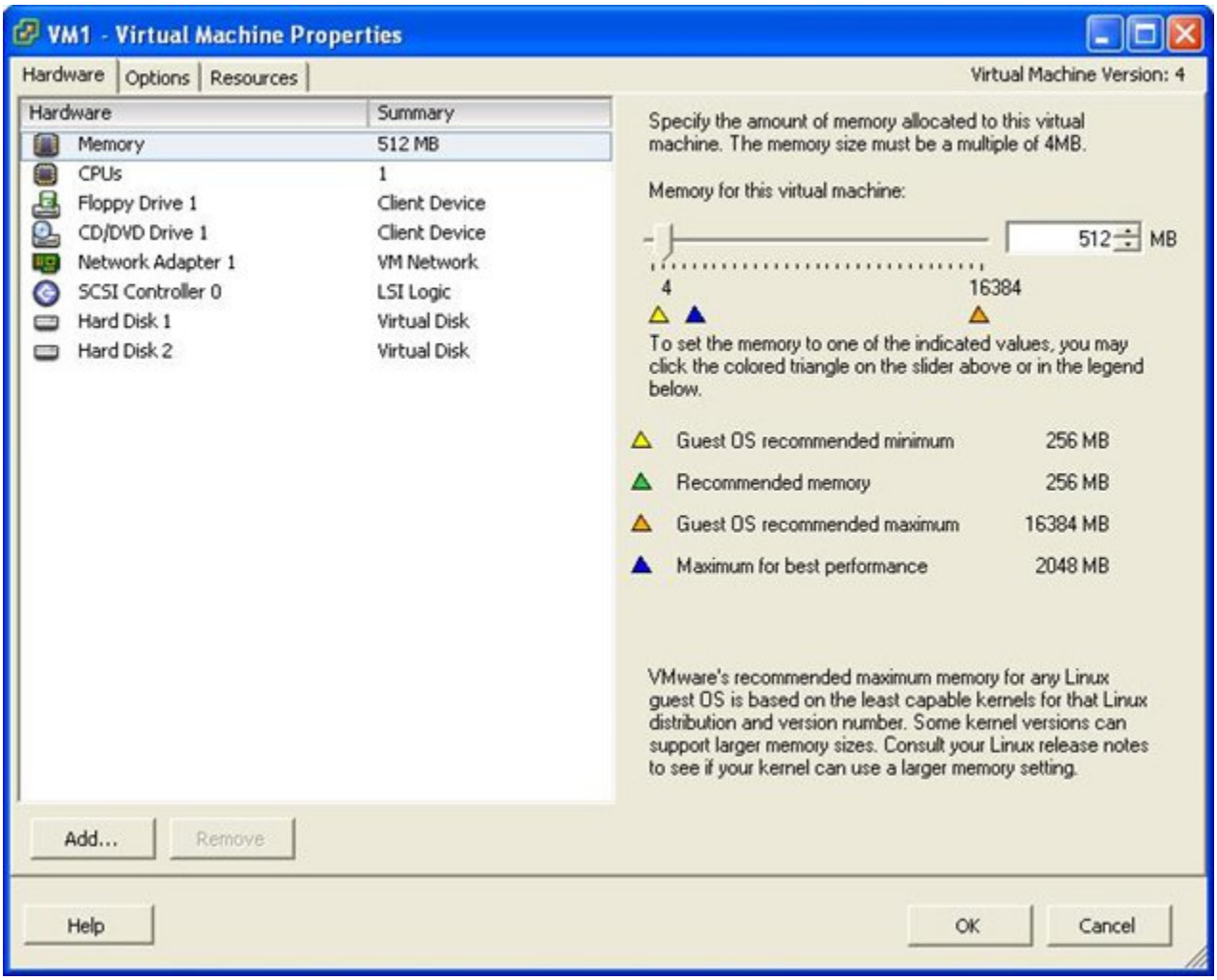
1. Download VirtualBox from its official website.

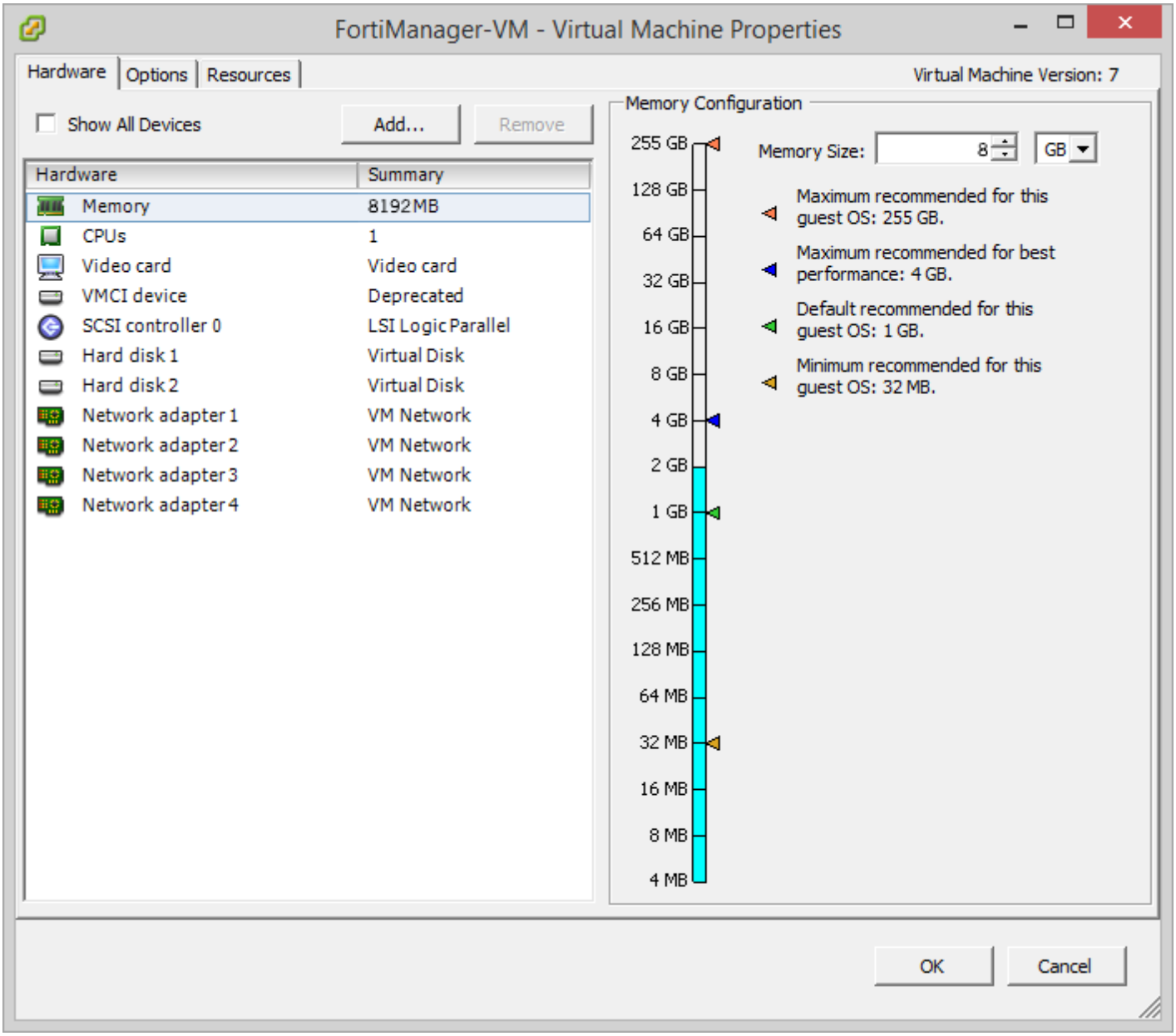
2. Install the software by following the setup wizard.
3. Open VirtualBox and click New.
4. Select OS Type: Linux Ubuntu (64-bit).
5. Allocate RAM (Minimum 2GB).
6. Create a virtual hard disk (Minimum 20GB).
7. Attach Ubuntu ISO file under Settings Storage.
8. Start the VM and complete Ubuntu installation.
9. Install Guest Additions for enhanced performance.

Part B: Installation and Configuration of VMware Workstation




New Virtual Machine Wizard
Select the Appropriate Configuration
How would you prefer to configure your new virtual machine?
Virtual machine configuration
Typical
Create a new virtual machine with the most common devices and configuration options.
Custom
Choose this option if you need to create a virtual machine with additional devices or specific configuration options.
< Back
Next >
Cancel


VM1 - Virtual Machine Properties
Hardware
Options
Resources
Virtual Machine Version: 4
Hardware
Summary
Memory
512 MB
CPUs
1
Floppy Drive 1
Client Device
CD/DVD Drive 1
Client Device
Network Adapter 1
VM Network
SCSI Controller 0
LSI Logic
Hard Disk 1
Virtual Disk
Hard Disk 2
Virtual Disk
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4MB.
Memory for this virtual machine:
512
MB
4
16384
To set the memory to one of the indicated values, you may click the colored triangle on the slider above or in the legend below.
Guest OS recommended minimum
256 MB
Recommended memory
256 MB
Guest OS recommended maximum
16384 MB
Maximum for best performance
2048 MB
VMware's recommended maximum memory for any Linux guest OS is based on the least capable kernels for that Linux distribution and version number. Some kernel versions can support larger memory sizes. Consult your Linux release notes to see if your kernel can use a larger memory setting.
Add...
Remove
Help
OK
Cancel



1. Download VMware Workstation Player.
2. Install and launch the software.
3. Click Create a New Virtual Machine.
4. Select Ubuntu ISO file.
5. Choose Linux Ubuntu 64-bit.
6. Allocate RAM (2–4GB) and Disk Space (20GB).
7. Finish setup and power on the VM.
8. Install VMware Tools for better integration.

Result

- Successfully installed and configured VirtualBox and VMware Workstation.
- Created and executed Ubuntu virtual machines on both hypervisors.
- Compared performance, interface, and configuration options of both tools.

Conclusion

The experiment demonstrated how two different Type-2 hypervisors can be installed and configured on a host operating system.

VirtualBox is user-friendly and open-source, suitable for beginners and educational purposes.

VMware Workstation offers better performance and advanced features, making it more suitable for professional and enterprise environments.

Thus, virtualization enables efficient resource utilization and allows multiple operating systems to run on a single physical machine.

Experiment No: 10

Installing and Operating a Guest Operating System Using a Hypervisor

Theory

A Hypervisor is software that allows you to create and run Virtual Machines (VMs). Each VM can run its own Guest Operating System (Guest OS) inside your main system (Host OS).

- Host OS Your main operating system (Windows/Linux/Mac)
- Guest OS OS installed inside virtual machine
- Hypervisor Software that manages virtual machines

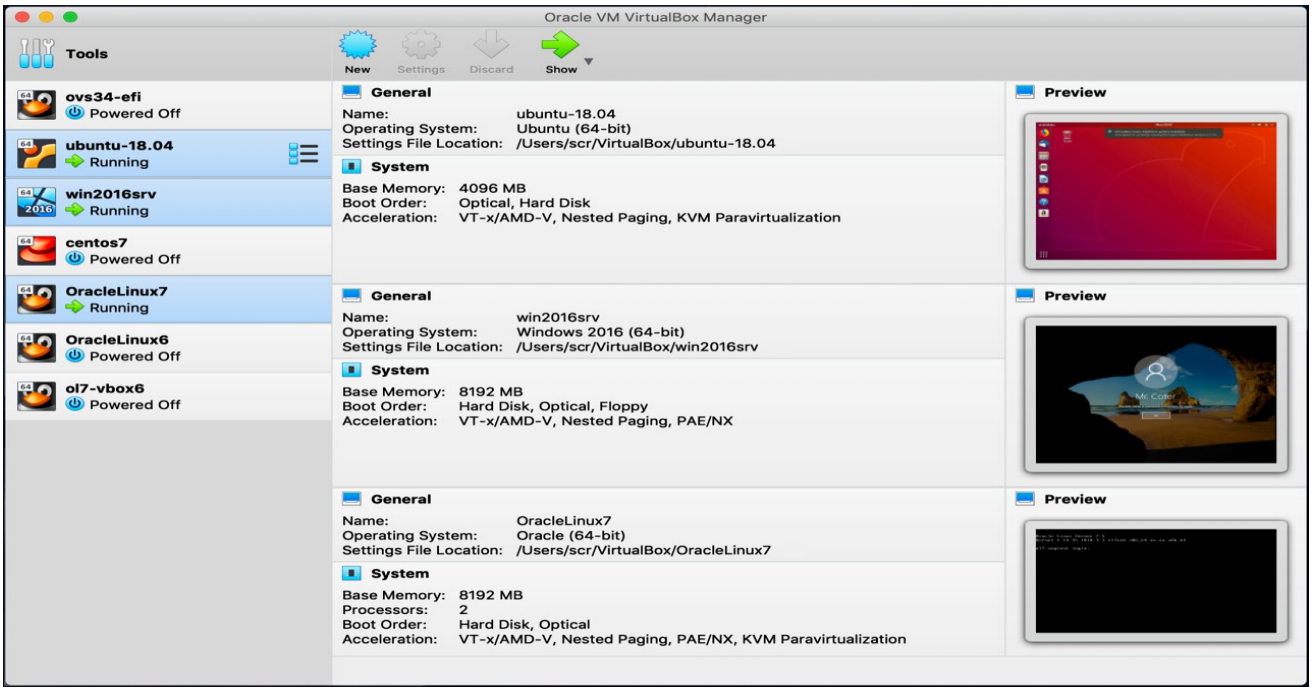
Examples:

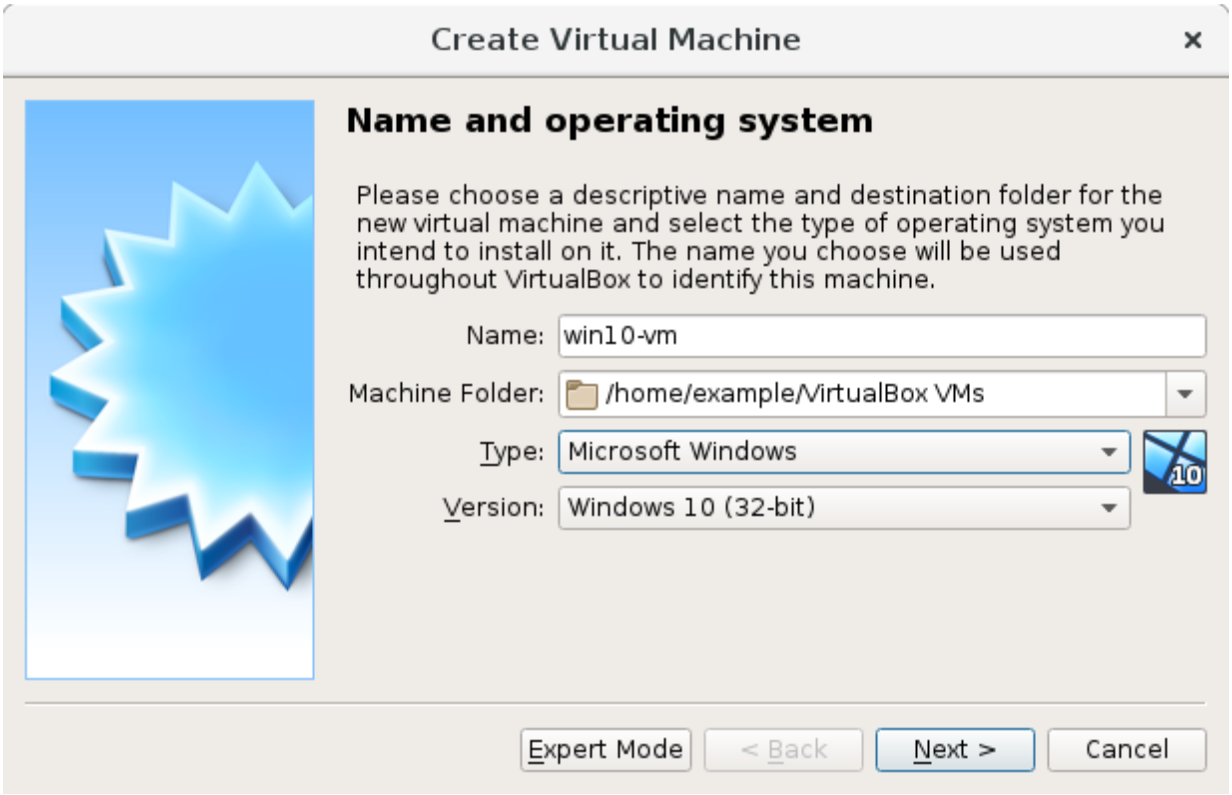
- Oracle VM VirtualBox
- VMware Workstation
- Microsoft Hyper-V

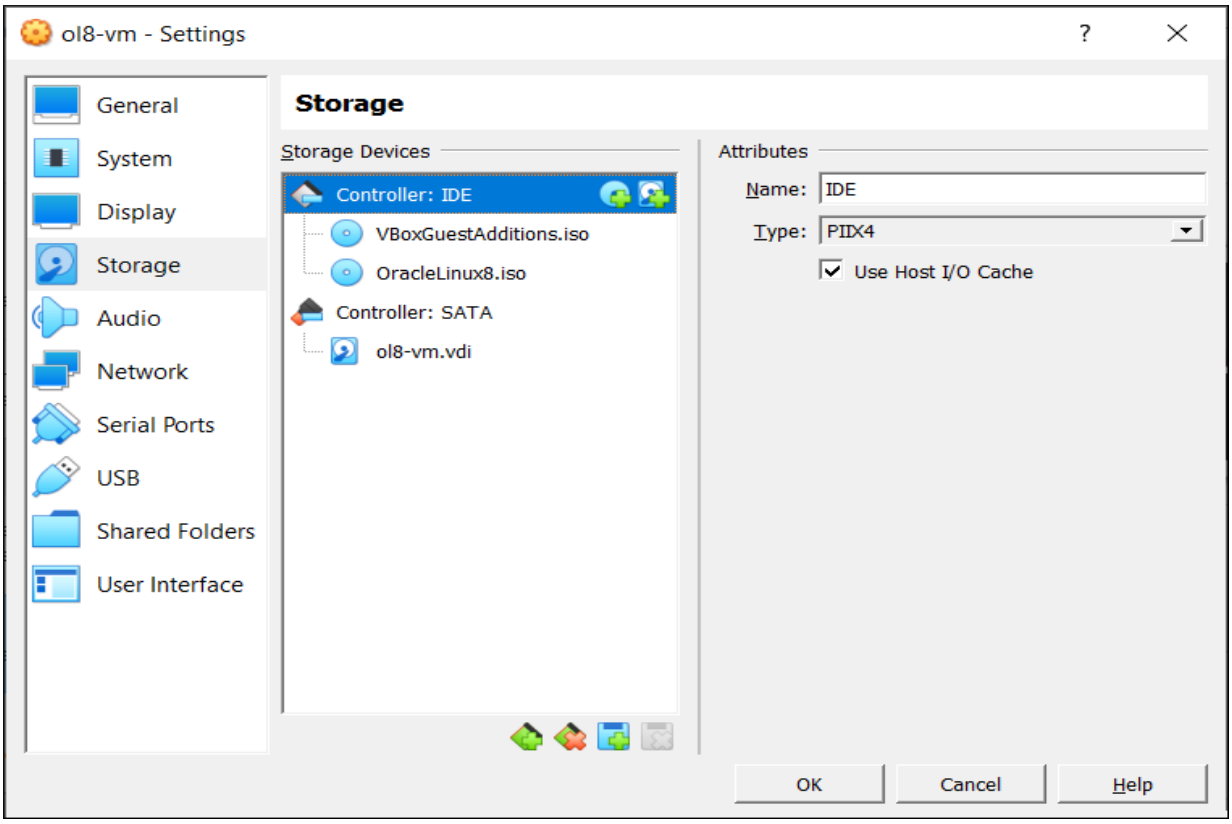
Aim

To install and operate a Guest Operating System using a hypervisor.

Popular Hypervisor Example – Oracle VM VirtualBox




Create Virtual Machine
Name and operating system
Please choose a descriptive name and destination folder for the new virtual machine and select the type of operating system you intend to install on it. The name you choose will be used throughout VirtualBox to identify this machine.
Name: win10-vm
Machine Folder: /home/example/VirtualBox VMs
Type: Microsoft Windows
Version: Windows 10 (32-bit)
Expert Mode
< Back
Next >
Cancel


ol8-vm - Settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Storage
Storage Devices
Controller: IDE
VBoxGuestAdditions.iso
OracleLinux8.iso
Controller: SATA
ol8-vm.vdi
Attributes
Name: IDE
Type: PIIX4
Use Host I/O Cache
OK
Cancel
Help

Requirements

- Host system (Windows/Linux)
- Hypervisor software (VirtualBox)
- Guest OS ISO file (Example: Ubuntu ISO file)
- Minimum 8GB RAM recommended

Step 1: Install Hypervisor

1. Download VirtualBox from official website.
2. Install it like normal software.
3. Restart system if required.

Step 2: Create New Virtual Machine

1. Open VirtualBox.
2. Click New.
3. Enter:
    - Name: Ubuntu
    - Type: Linux
    - Version: Ubuntu (64-bit)
4. Click Next.

Step 3: Allocate Resources

RAM Allocation

- Minimum: 2 GB
- Recommended: 4 GB

Create Virtual Hard Disk

- Choose VDI
- Select Dynamically allocated
- Size: 20 GB or more

Click Create.

Step 4: Attach ISO File

1. Go to Settings Storage

2. Under Controller IDE, click Empty
3. Choose Ubuntu ISO file
4. Click OK

Step 5: Start Virtual Machine

1. Click Start
2. Ubuntu installer screen appears
3. Select:
   - o Install Ubuntu
   - o Language
   - o Keyboard layout
4. Choose:
   - o Normal Installation
   - o Erase disk (only inside VM, safe)
5. Create username & password
6. Click Install

Installation takes 10–20 minutes.

Step 6: Operating the Guest OS

After installation:

- Login using username & password
- Open Terminal
- Install software
- Create files
- Run applications

Example commands:

```
ls
pwd
mkdir test
cd test
```

Installing Guest Additions (Important)

1. Go to Devices Insert Guest Additions CD Image
2. Install it inside Ubuntu

3. Restart VM

Benefits:

- Full screen mode
- Clipboard sharing
- Better graphics
- Drag & drop support

Types of Hypervisors

Type	Description	Example
Type 1	Runs directly on hardware	Hyper-V
Type 2	Runs on host OS	VirtualBox, VMware

Advantages of Using Hypervisor

Run multiple OS on one machine

Safe testing environment

Software development & testing

No need for dual boot

Snapshot & restore feature

Conclusion

Installing and operating a Guest Operating System using a hypervisor allows users to run multiple operating systems on a single physical machine. It provides a secure, flexible, and efficient environment for testing, development, and learning without affecting the host system.

Experiment no: 11

Creating and managing a repository on github

Aim

To create and manage a repository using GitHub and perform basic version control operations.

Objectives

1. To create a new repository on GitHub.
2. To clone the repository to the local system.
3. To add, commit, and push files to GitHub.
4. To create and manage branches.
5. To understand basic Git commands.

Requirements

- Computer System
- Internet Connection
- Git Installed
- GitHub Account

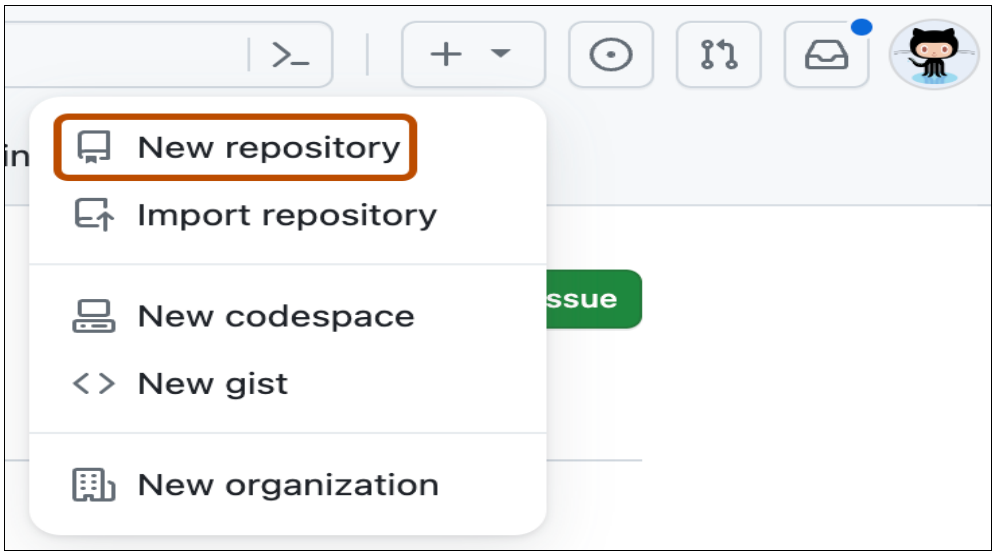
Theory

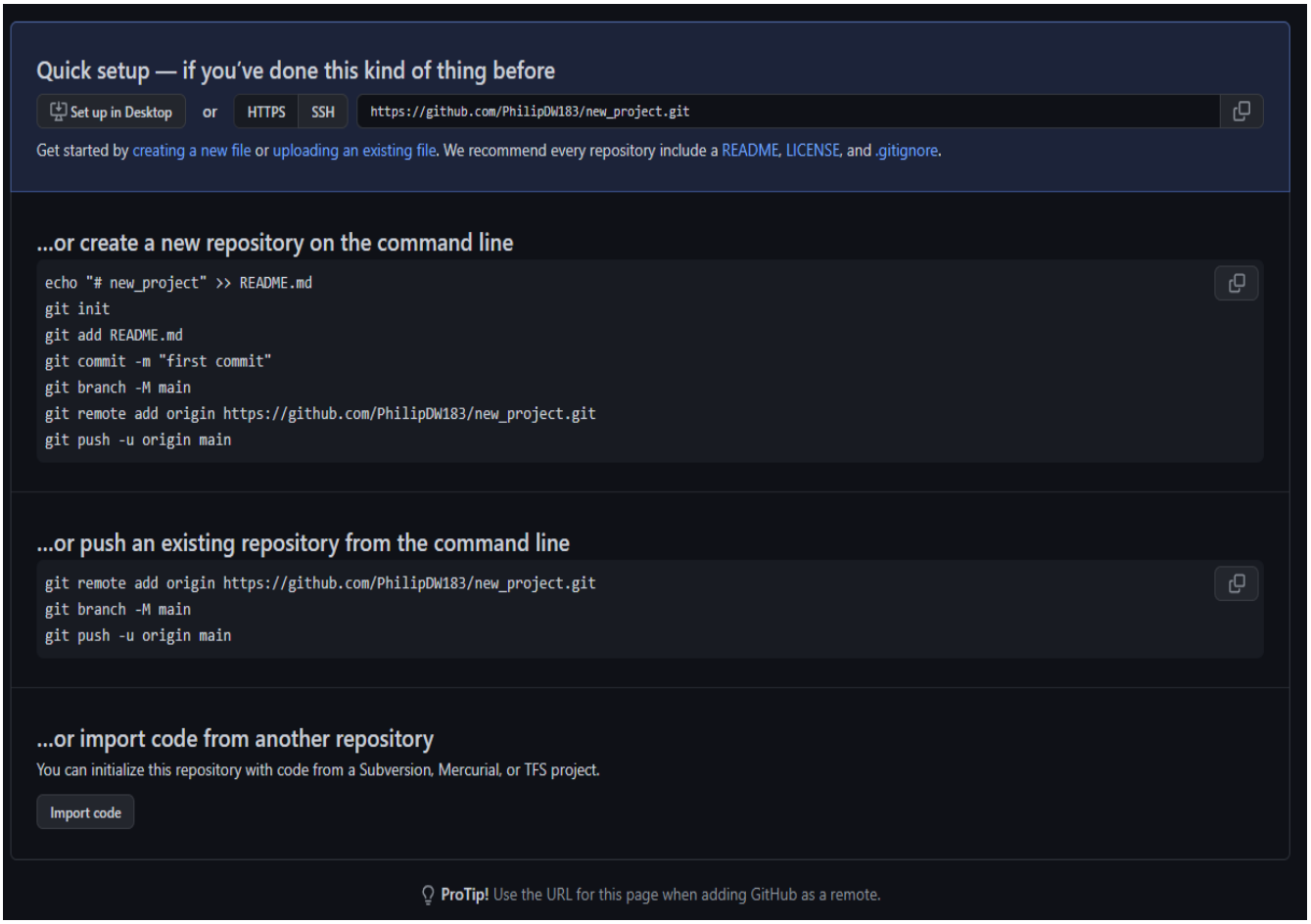
GitHub is a cloud-based platform used for version control and collaboration. It allows developers to store, manage, and track changes in their projects using Git.

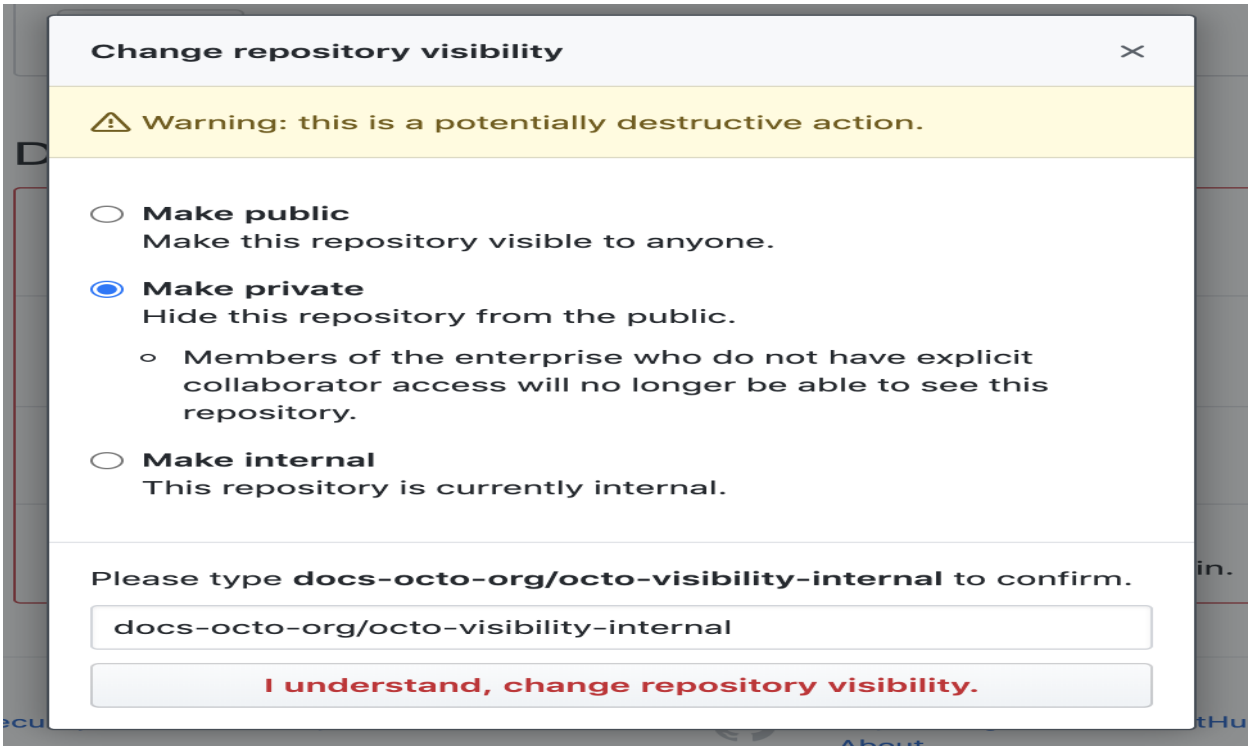
A repository is a storage location where project files and their revision history are maintained. Git provides commands like add, commit, push, pull, branch, and merge to manage changes efficiently.

Procedure

Step 1: Create Repository on GitHub


New repository
Import repository
New codespace
New gist
New organization


Quick setup — if you've done this kind of thing before
Set up in Desktop
or
HTTPS
SSH
https://github.com/PhilipDW183/new_project.git
Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.
...or create a new repository on the command line
echo "# new_project" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/PhilipDW183/new_project.git
git push -u origin main
...or push an existing repository from the command line
git remote add origin https://github.com/PhilipDW183/new_project.git
git branch -M main
git push -u origin main
...or import code from another repository
You can initialize this repository with code from a Subversion, Mercurial, or TFS project.
Import code
ProTip! Use the URL for this page when adding GitHub as a remote.



1. Open https://github.com
2. Login to your account.
3. Click on "+" icon New repository.
4. Enter repository name.
5. Select Public or Private.
6. Click Create repository.

Step 2: Clone Repository

Copy the repository URL and run:

```
git clone https://github.com/username/repository-name.git
cd repository-name
```

Step 3: Add and Commit Files

```
git add .
git commit -m "Initial Commit"
git push origin main
```

Step 4: Create and Manage Branch

```
git checkout -b new-branch
git checkout main
git merge new-branch
git branch -d new-branch
```

Step 5: Pull Latest Changes

```
git pull origin main
```

Observation

- Repository successfully created on GitHub.
- Files uploaded using Git commands.
- Branch created and merged successfully.
- Version history maintained properly.

Result

The repository was successfully created and managed using GitHub. All Git operations such as clone, add, commit, push, pull, branch, and merge were performed successfully.

Conclusion

GitHub provides an efficient platform for project version control and collaboration. By using Git commands, users can manage project files, track changes, and work effectively in teams.

Experiment: 12

Collaborative Development Using Branching and Pull Requests

Aim

To understand and implement collaborative development using branching and pull requests in Git and GitHub.

Objectives

- To create and manage branches in Git.
- To perform commits and push changes to GitHub.
- To create and review pull requests.
- To merge branches and resolve conflicts.

Software / Tools Required

- Git (installed on system)
- GitHub Account
- VS Code / Any Code Editor
- Internet Connection

Theory

Collaborative development allows multiple developers to work on the same project simultaneously.

- Branching enables independent development without affecting the main code.
- Pull Requests (PRs) allow code review before merging changes into the main branch.

This workflow improves code quality, avoids conflicts, and ensures proper version control.

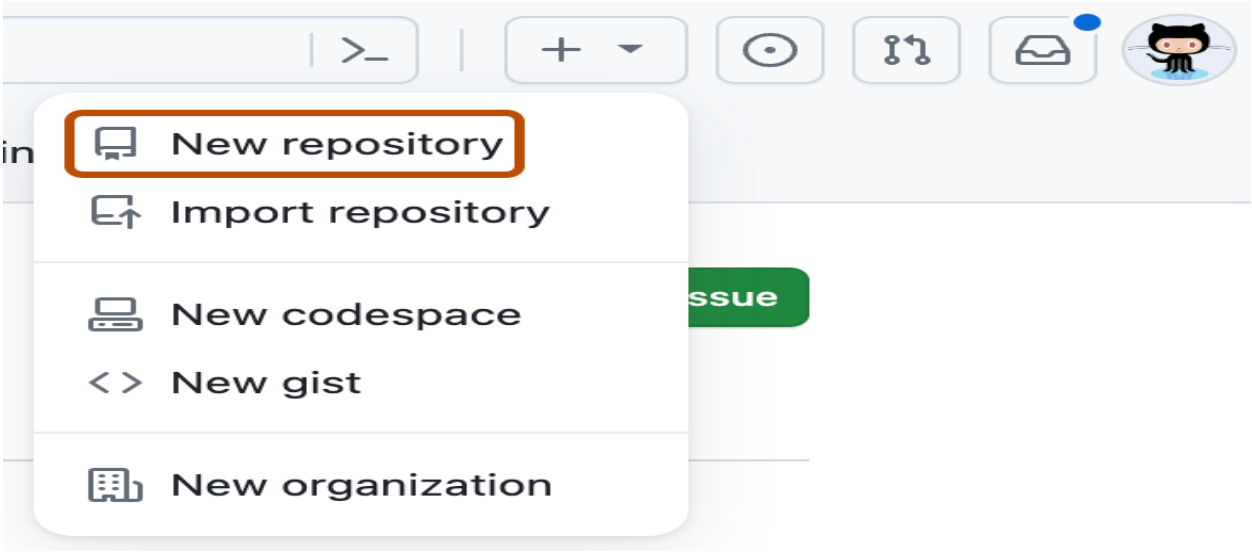
Procedure (Stepwise Practical Execution)

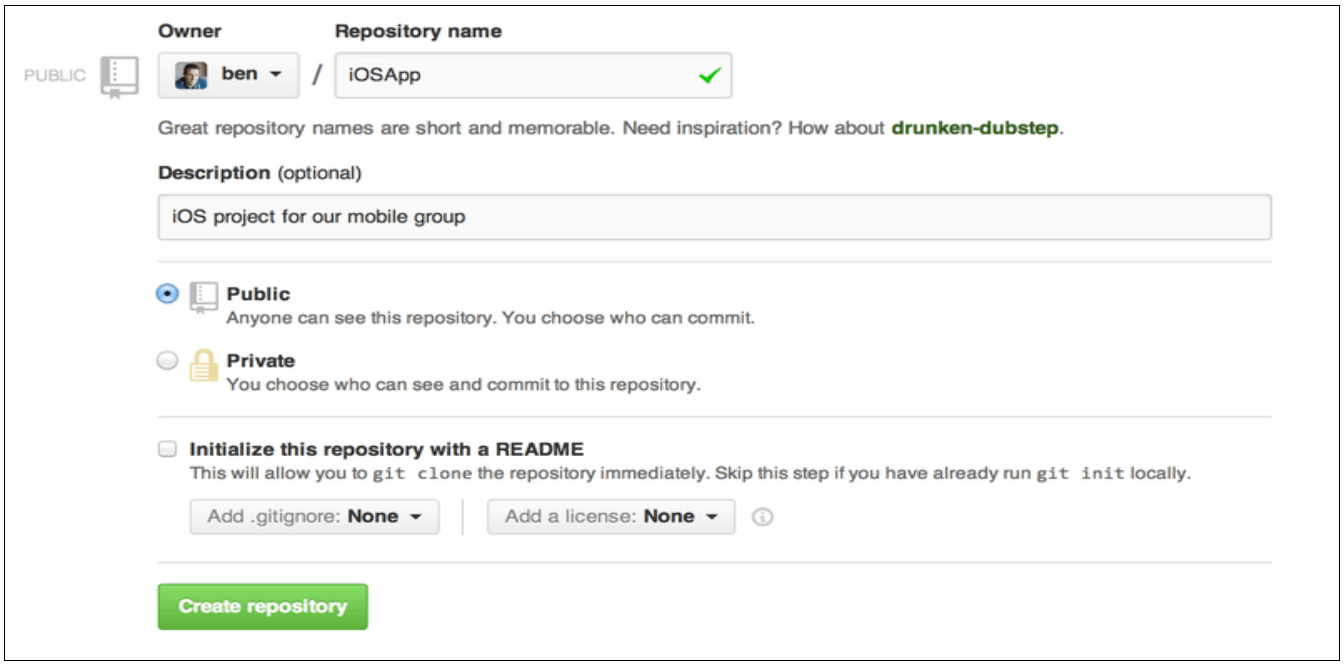
Step 1: Install Git

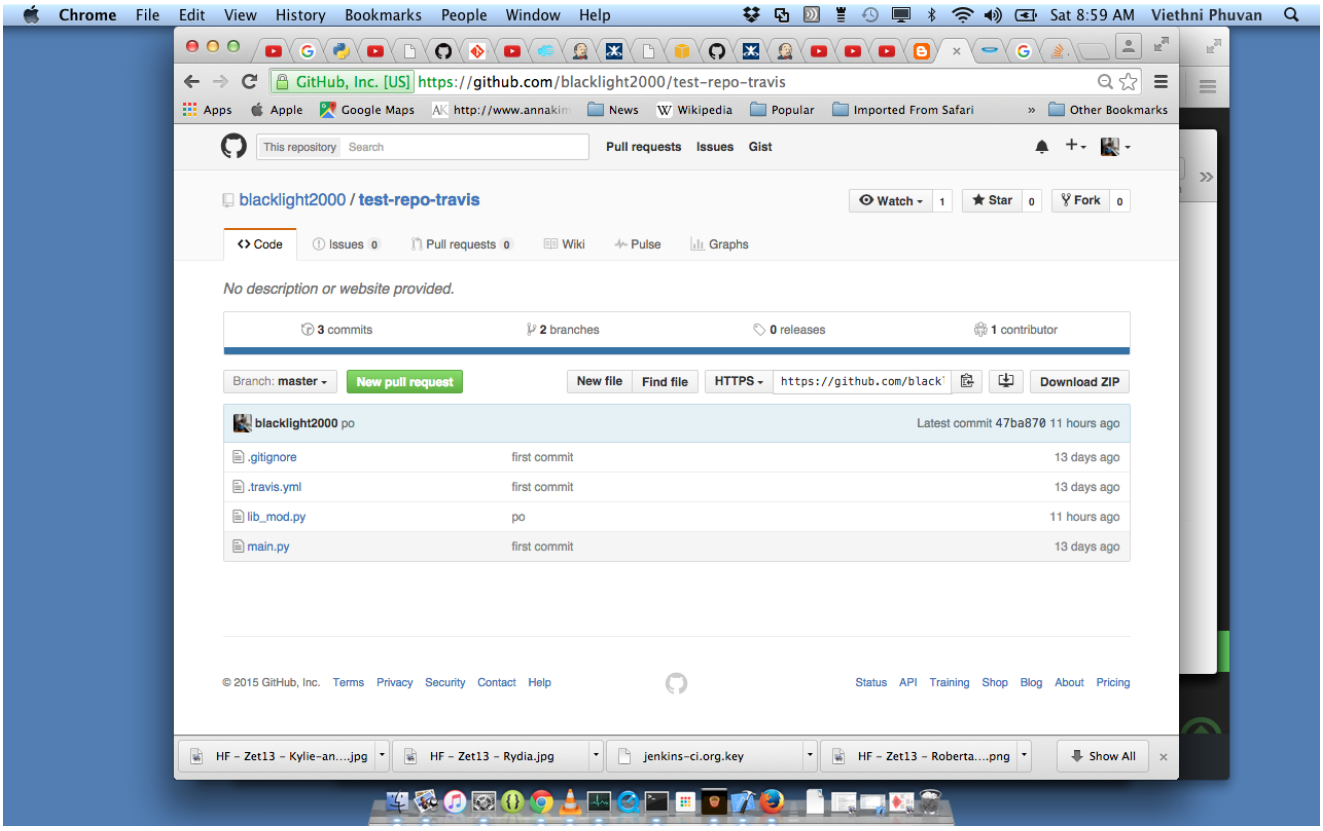
- Download Git from official website.
- Install with default settings.
- Verify installation:

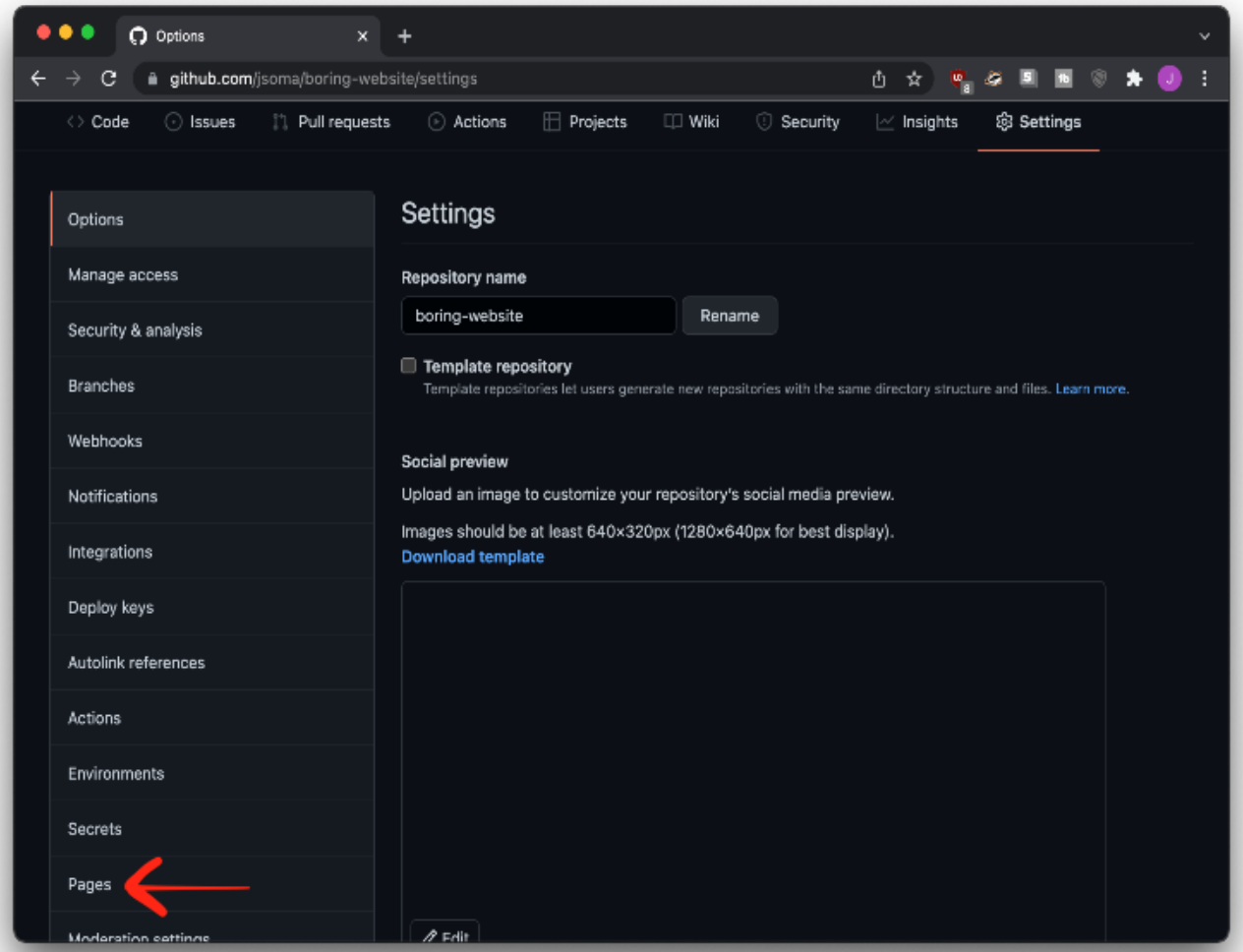
```
git --version
```

Step 2: Create a New Repository on GitHub






Chrome File Edit View History Bookmarks People Window Help
Sat 8:59 AM Viethni Phuvan
GitHub, Inc. [US] https://github.com/blacklight2000/test-repo-travis
Apps Apple Google Maps http://www.annakim News Wikipedia Popular Imported From Safari Other Bookmarks
This repository Search
Pull requests Issues Gist
blacklight2000 / test-repo-travis
Watch 1 Star 0 Fork 0
Code Issues 0 Pull requests 0 Wiki Pulse Graphs
No description or website provided.
3 commits 2 branches 0 releases 1 contributor
Branch: master New pull request New file Find file HTTPS https://github.com/black Download ZIP
blacklight2000 po Latest commit 47ba870 11 hours ago
.gitignore first commit 13 days ago
.travis.yml first commit 13 days ago
lib_mod.py po 11 hours ago
main.py first commit 13 days ago
© 2015 GitHub, Inc. Terms Privacy Security Contact Help
Status API Training Shop Blog About Pricing
HF – Zet13 – Kylie-an....jpg
HF – Zet13 – Rydia.jpg
jenkins-ci.org.key
HF – Zet13 – Roberta....png
Show All



- Login to GitHub
- Click New Repository
- Enter repository name
- Click Create Repository

Step 3: Clone Repository to Local System

```
git clone <repository-url>
cd repository-name
```

Step 4: Create a New Branch

```
git branch feature-login
git checkout feature-login
```

OR

```
git checkout -b feature-login
```

Verify branch:

git branch

Step 5: Make Changes

- Create a new file (example: login.html)
- Add some content
- Save the file

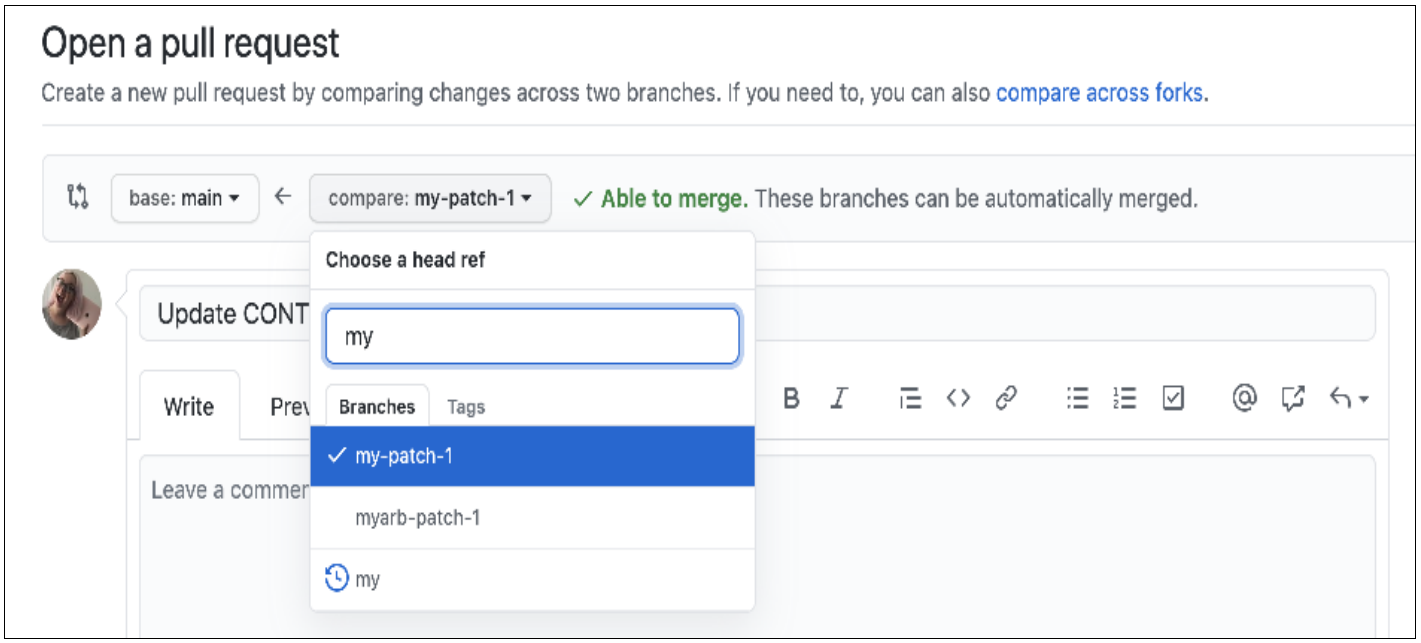
Step 6: Add and Commit Changes

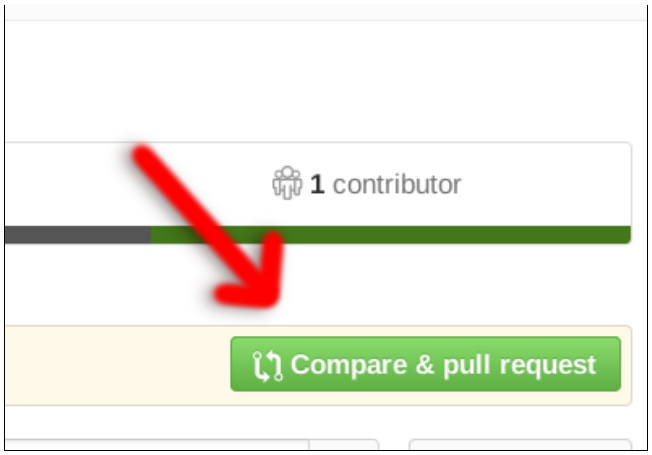
```
git add .
git commit -m "Added login page feature"
```

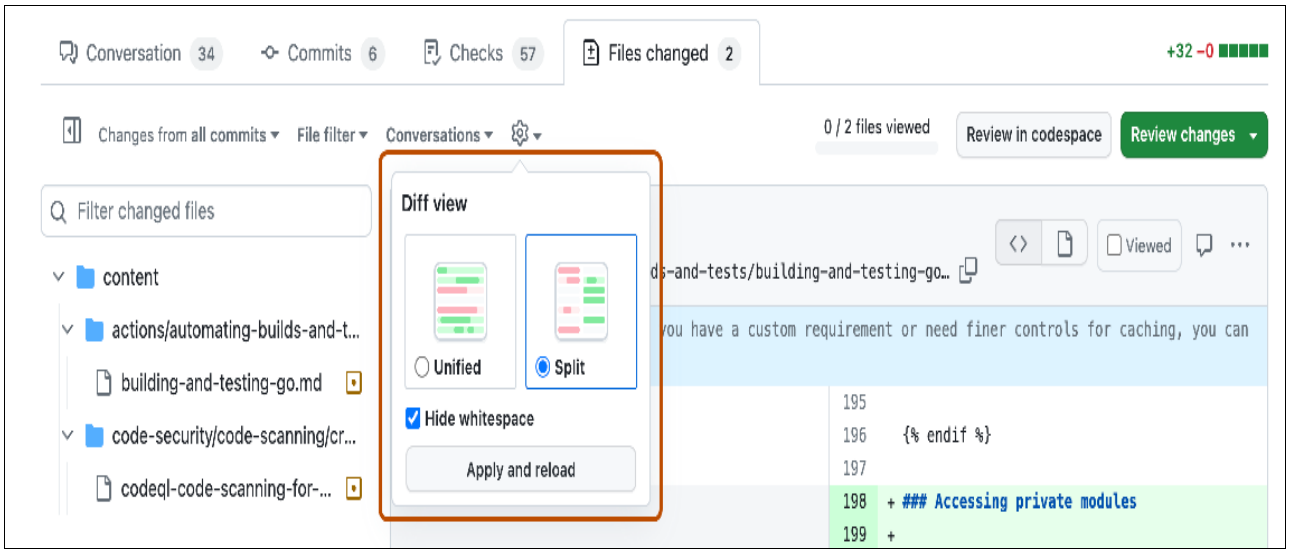
Step 7: Push Branch to GitHub

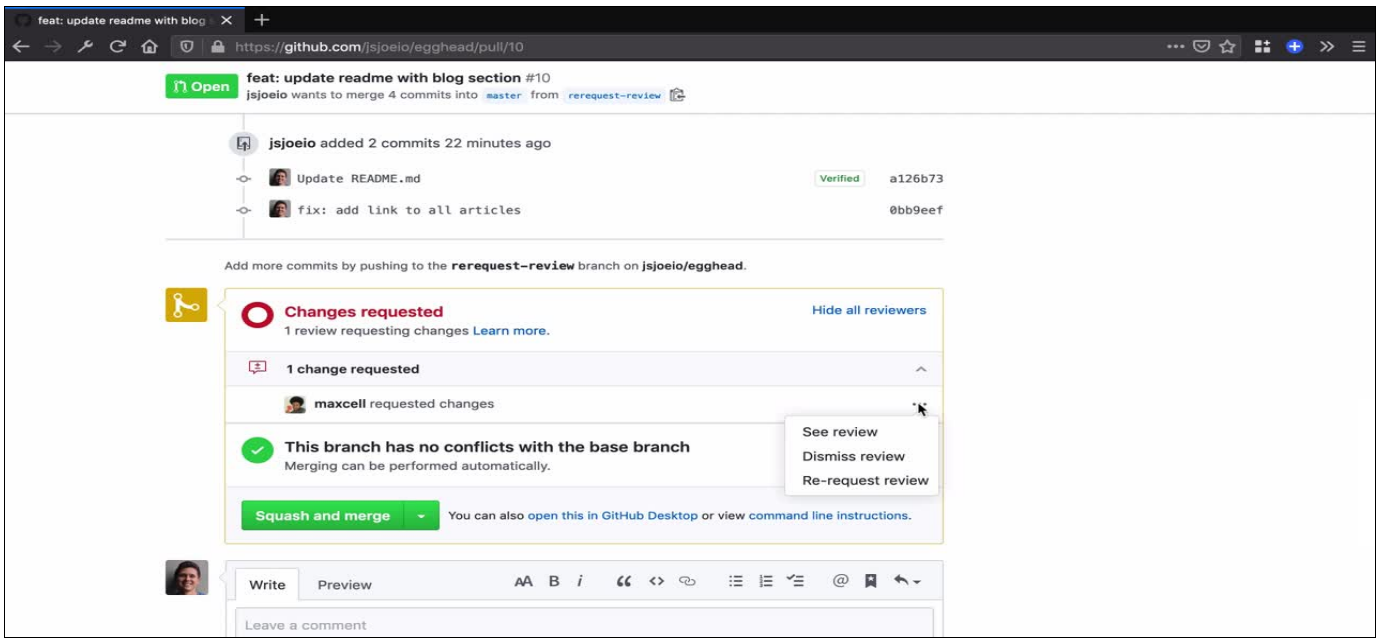
```
git push origin feature-login
```

Step 8: Create Pull Request on GitHub









- Go to repository on GitHub
- Click Compare & Pull Request
- Add title and description
- Click Create Pull Request

Step 9: Review and Merge Pull Request

- Review changes
- Click Merge Pull Request
- Confirm Merge
- Delete feature branch (optional)

Step 10: Pull Latest Changes to Local

```
git checkout main
git pull origin main
```

Result

The experiment was successfully completed. A new branch was created, changes were committed and pushed, a pull request was created, reviewed, and merged into the main branch.

Conclusion

Collaborative development using branching and pull requests ensures organized teamwork, maintains code quality, and allows safe integration of new features into the main project.